

Drug Dosing Considerations in Patients With Acute Kidney Injury and Chronic Kidney Disease

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KEY POINTS

- Chronic kidney disease (CKD) results in alterations in the absorption, volume of distribution (V_D), metabolism, and elimination of most drugs.
- The V_D of many drugs is increased in the presence of acute and CKD due to volume expansion and/or reduced protein binding.
- In addition to the expected decrement in kidney clearance, nonkidney clearance (i.e., gastrointestinal and hepatic drug metabolism) of several drugs is also reduced in CKD patients.
- Individualization of a drug dosage regimen for a patient with reduced kidney function is based on the pharmacodynamic/pharmacokinetic characteristics of the drug, the patient's degree of residual kidney function, and his or her overall clinical condition.
- Patient covariates, disease covariates, and pharmacogenetics all contribute to variability in drug disposition.
- The drug dosing guidelines for CKD patients in many drug information resources are highly variable, and many have not been optimized for clinical use.
- The effect of hemodialysis, peritoneal dialysis, and continuous kidney replacement therapy on drug elimination is dependent on the characteristics of the drug and the dialysis prescription.
- Prospective monitoring of serum concentrations is often warranted, especially for narrow therapeutic index drugs.

DRUG DOSING CONSIDERATIONS

The goals of drug dose adjustment to an individual patient are twofold: Obtain the same pharmacodynamic response when pharmacokinetics change, and minimize the risk of drug toxicity when organ function fails.

It is estimated that 10% to 15% of the global population has chronic kidney disease (CKD).¹⁻³ The prevalence varies widely across the world, partly because of heterogeneity of the laboratory methods used to detect CKD; environmental factors; public health policies; and genetics.³ The incidence of CKD has more than doubled in the past 20 years in adults older than 65 years.⁴ This is due to age-related reductions in

kidney function, multiple medical comorbidities, increases in the prevalence of older individuals who have survived cardiovascular disease, and increased use of medications that alter kidney function, among others.⁵ CKD can affect multiple organ systems, and the associated pathophysiologic changes can profoundly alter the pharmacokinetics and pharmacodynamics of many drugs.⁵⁻⁷ Clinicians must assess kidney function and consider how it alters the disposition of drugs and their active or toxic metabolites to optimize the health outcomes of their patients.

The epidemiology of acute kidney injury (AKI) varies widely due to the variability in the criteria used for diagnosis.⁸ AKI has a frequency of 2% in hospitalized patients, but its prevalence among the critically ill is up to 30 times greater.⁹⁻¹³ The number of patients with AKI has increased in the past 10 years in most regions of the world.¹³ The development of some degree of CKD and/or need for chronic kidney replacement therapy (KRT) are increasingly common among some survivors of AKI.¹⁴ The widespread use of kidney replacement therapies in patients with AKI (e.g., continuous replacement therapy [CRRT]) and end-stage kidney failure (ESKF; hemodialysis [HD] or hemodiafiltration) mandates an understanding of their impact on drug disposition in these high-risk patients.^{15,16} Advances in dialyzer technology and delivery systems have resulted in enhanced efficiency of dialysis treatments. Unfortunately, these advances have rarely been accompanied by reevaluations of the influence of dialysis on drug disposition.^{6,7,17}

Medications that are predominantly eliminated unchanged by the kidney may accumulate in patients with CKD, which can increase the risk of adverse effects. As a general concept, if 30% or more of a drug is eliminated unchanged in the urine, it will have a high likelihood of requiring dose adjustment in CKD, especially in stages 3 to 5.^{5,7,18} The pharmacokinetics of drugs with a fractional extraction [FE] <30% may also be affected and thus require dose adjustment. In fact, 32.2% of drugs approved in the United States between 1998 and 2010 included a dose-adjustment recommendation for CKD in the product labeling.¹⁹ Despite an increasing number of studies considering kidney function by the pharmaceutical and biotechnology industry and improvements in approved product labeling language, challenges remain for dose adjustments in CKD, which are especially important for oncology and antiretroviral agents.²⁰

Although P pharmacokinetic and pharmacodynamic studies in patients with CKD are being performed for an increasing proportion of drugs, for >70% of drugs released in the United States, pharmacokinetic and pharmacodynamic study results have not been published in the past decade.¹⁹ Regulatory authorities have not yet responded to the challenge of ensuring robust data for patients with impaired kidney function by requiring a pharmacokinetic and pharmacodynamic investigation plan as a routine part of drug development.⁶ Furthermore, patients with moderate to severe CKD are typically excluded from participation in major safety and efficacy studies required for drug registration. Therefore the initial dosage recommendations cannot be evaluated in those with CKD or, at times, conditions for which they are intended. If there is no official dosage regimen recommendation in

the product labeling, an adjustment may be calculated on the basis of the drug's fractional excretion and the ratio of the patient's estimated creatinine clearance (eCrCl) or estimated glomerular filtration rate (eGFR) relative to an age and gender normal value.^{21,22}

The pharmaceutical industry began to investigate the relationship of kidney function to the pharmacokinetics and pharmacodynamics of the drugs they had in development in the 1970s and this continues to evolve.²³ However, much of the data on the pharmacokinetics of drugs in patients with kidney disease derives from clinician-initiated, postmarketing studies. Significant differences exist with respect to the means of assessment of kidney function and classification of the degree of impaired kidney function in the available studies.¹⁸ Studies conducted to meet global regulatory expectations have resulted in the generation of conflicting results for some drugs.²⁴⁻²⁷ Despite the availability of this extensive collection of published evidence, dosing errors in CKD patients still occur at an alarming rate.^{20,28-30} The expanded use of electronic medical records has not replaced the need for clinician proactivity to optimize the use of medications in CKD patients. Studies have shown that up to 85% of the medications ordered had nephrotoxic potential and >20% of the drugs ordered were not dose adjusted for the patient's kidney function.^{28,31-33} Thus the optimization of therapy for patients with AKI and CKD is ultimately dependent on the clinicians' utilization of the data that are available.

In this chapter, the influence of AKI and CKD on drug pharmacokinetic properties is characterized, and a guide for individualizing drug therapy in patients with AKI and CKD is presented. We also present dosage recommendations for many commonly used drugs. The role of pharmacokinetic measures alone or in combination with pharmacodynamics, as well as pharmacogenetic testing in drug dosage regimen design, are all discussed. The impact on drug disposition through maintenance dialysis for ESKF or continuous KRT (CKRT) for patients with AKI is discussed. Dosing recommendations for many critical drugs are presented.

EFFECTS OF ACUTE KIDNEY INJURY AND CHRONIC KIDNEY DISEASE ON DRUG DISPOSITION

Pharmacokinetics describes the time course of drug absorption, distribution, metabolism, and elimination. Pharmacodynamics provides a characterization of the complex interaction of drug concentrations, receptor-drug interactions, mechanism of action, and clinical factors, such as concurrent diseases and degree of organ dysfunction on patients' response to drug therapy. The combination of pharmacokinetic and pharmacodynamic drug characteristics allows clinicians with foundational information to make rational prescribing decisions.

When given intravenously (IV), a rapid decrease in the plasma concentration follows an initial high drug concentration. This decrease occurs as the drug distributes from the plasma into the extravascular space and beyond. During the terminal elimination phase, drug concentrations in plasma are in equilibrium with concentrations in body tissues (Fig. 56.1). The rate and extent of drug absorption

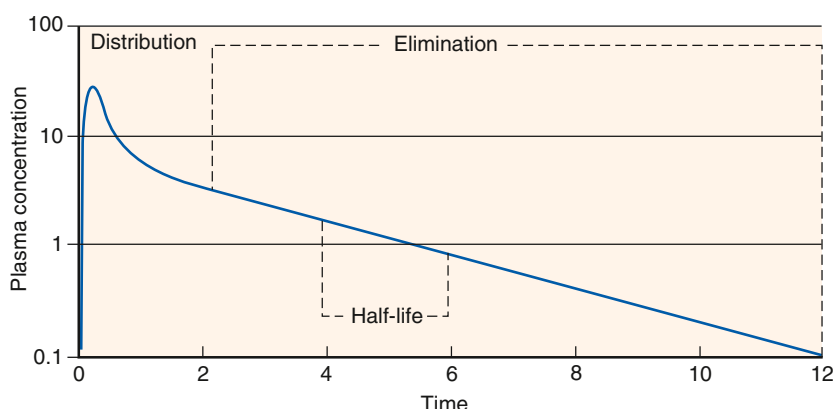


Fig. 56.1 Distribution and elimination of a drug after intravenous administration.

and distribution and the rate of drug elimination may be ascertained by mathematic analysis of the serum or plasma concentration data collected at appropriate time intervals. The terminal elimination half-life of a drug is the time required for the plasma concentration to decline by 50%; this can be determined from the slope during the elimination phase of the plot of serum or plasma drug concentrations versus time after drug ingestion or injected. By comparing pharmacokinetic data from patients with normal kidney function with pharmacokinetic data from patients with impaired kidney function, rational drug dosing regimens may be proposed.^{21,22}

ABSORPTION

Drugs given IV enter the central circulation directly and generally have a rapid onset of action. The absolute bioavailability is determined by comparing the area under the serum/plasma concentration–time curve (AUC) after oral administration with that observed after IV administration. Drugs given by other routes must first pass through important organs of elimination (e.g., liver) before entering the systemic circulation; thus a smaller proportion of the drug becomes available at the site of drug action. Even drugs given IV and by inhalation must pass through the lungs before reaching arterial blood. Similar to other organs, the lungs remove substantial amounts of some agents (e.g., IV administered adenosine). For drugs administered orally, the rate and extent of gastrointestinal (GI) absorption are important considerations. Absorption can be characterized by determining the maximum attained serum or plasma concentration (C_{max}), as well as the time after ingestion when the C_{max} was observed (T_{max}). Differences in these two parameters (C_{max} and T_{max}) among patient groups were historically considered evidence of altered GI absorption, but in many cases the absolute bioavailability was unchanged.³⁴ The bioavailability of a drug depends on the extent of metabolism during its first pass through the intestinal wall and liver before reaching the systemic circulation. When this AUC-derived measure of bioavailability was assessed, there were few drugs shown to be affected by the presence of CKD or AKI.^{7,35}

First-pass biotransformation may also occur in the gut; bioflavonoids in grapefruit juice can inhibit cytochrome P450 (CYP) 3A4 and noncompetitively inhibit the

metabolism of drugs metabolized by this enzyme. The grapefruit juice–CYP3A4 interaction was first noted with the calcium channel blocker felodipine.³⁶ This interaction also increases the bioavailability of cyclosporine by as much as 20%.³⁷ A wide variety of other drugs are similarly affected, including several medications used for depression and anxiety (e.g., selective serotonin reuptake inhibitors and serotonin–norepinephrine reuptake inhibitors) and statins.³⁸ Herbal medicine (e.g., hypericin, one of the constituents of St. John’s wort) can activate the adenosine triphosphate–binding cassette transporter or P-glycoprotein (multidrug resistance) transporter (e.g., cyclosporine) in gut mucosa, leading to reduced drug absorption.³⁹

Although GI symptoms are common in patients with ESKF, little specific information about gastrointestinal function is available. The salivary concentration of urea increases when urea accumulates in plasma. Ammonia forms from urea in the presence of gastric urease and buffers gastric acid, increasing gastric pH. The ammonia is absorbed and converted to urea again by the liver. The gastric alkalinizing effect of this internal urea–ammonia cycle decreases the absorption of drugs that are best absorbed in an acidic environment. Drug malabsorption may be further aggravated by the increased use of various therapies to reduce gastric acidity and/or reduce phosphate absorption, especially in patients who are dialysis dependent.^{40,41} The resultant chelation and formation of nonabsorbable complexes reduce the bioavailability of some drugs, including several antibiotics and digoxin.

The processes of GI drug absorption are complex, may be saturable and dose dependent, and are more variable in patients with ESKF than in those with normal kidney function. Gastroparesis, commonly observed in patients with diabetes mellitus, many of whom also have CKD, prolongs gastric emptying and delays drug absorption (i.e., T_{max} is observed to be delayed). Conversely, diarrhea decreases gut transit time (T_{max} is shortened and diminishes drug absorption by the small bowel). Gut mucosal integrity becomes impaired across the spectrum of CKD, as evidenced by increasing levels of circulating translocated endotoxins.⁴²

DISTRIBUTION

The volume of distribution (V_D) of a drug does not necessarily correspond to a specific anatomic space. Rather,

the V_D is a mathematic construct based on the plasma concentration achieved following the IV administration of a given dose of a drug. Agents that are highly protein bound and those that are water soluble tend to be restricted to the vascular compartment and extracellular fluid (ECF) space, respectively, and thus have volumes of distribution less than 0.20 L/kg. Highly lipid-soluble drugs and those extensively bound to tissues often exhibit volumes of distribution in excess of 1 L/kg. The drug distribution volume of highly water-soluble or protein-bound drugs may be increased in patients with AKI or CKD if edema and/or ascites are present (Table 56.1).^{5,7,35,43} Drug distribution is one of the most important and complicated factors to quantify in patients with AKI. There is a fine balance between detrimental fluid overload and adequate hydration to preserve and optimize perfusion and function. Critically ill patients should be managed in a slightly negative fluid balance after initial adequate fluid resuscitation has been achieved.⁴⁴⁻⁴⁶ If patients are volume expanded, the administration of the usual doses of many drugs will result in inadequately low plasma concentrations.

The distribution volume of drugs may be altered by fluid removal during dialysis. Changes in body cell mass (nonfat, nonwater, and nonbone mineral mass) commonly occur over time in patients on dialysis, often accompanied by sarcopenia.⁴⁷ Failure to detect a reduction in body cell mass may lead to inappropriate maintenance of the same dry weight target and drug dosage regimen, despite a real increase in total body water, and thus the distribution volume of many drugs.⁴⁸ Finally, the method used to calculate the V_D may be influenced by impaired kidney function. There are three commonly used V_D terms: V_D of the central compartment (V_c), V_D of the terminal phase (V_β), and V_D at steady state (V_{ss}). The V_c for many drugs approximates ECF volume and thus may be increased or decreased by acute changes. Oliguric acute kidney failure is often accompanied by fluid overload and a resultant increased V_c for many drugs. The V_β represents the proportionality constant between plasma concentrations in the terminal elimination phase and the amount of drug remaining in the body. V_β is affected by distribution characteristics of the drug and by the terminal elimination rate constant. V_β and V_{ss} will often be similar in magnitude, with V_β being slightly larger.⁴⁹

Alterations of plasma protein binding in patients with CKD can also affect drug distribution and elimination. The V_D of a drug, the quantity of unbound drug available for action, and the degree to which the agent is eliminated by hepatic or kidney excretion are all influenced by protein binding. Drugs that are protein bound attach reversibly to albumin or α_1 -glycoprotein in plasma (Fig. 56.2). Whereas organic acids bind to a single binding site, organic bases probably have multiple sites of attachment.⁵⁰⁻⁵³

A combination of decreased serum albumin concentration and reduction in albumin affinity for the drug reduces protein binding in dialysis-dependent patients. The latter reflects, in part, the effect of protein-bound organic acids such as hippuric acid, indoxyl sulfate, and 3-carboxy-4-methyl-5-propyl-2-furanpropionic acid, which accumulate in advanced CKD, thereby decreasing the protein binding of many acidic drugs.^{53,54} This impaired plasma protein binding increases the unbound fraction of several acidic drugs in CKD, the magnitude of which correlates directly

Table 56.1 Volume of Distribution of Selected Drugs in Patients With Normal Kidney Function and Those on Dialysis

Drug	Normal (L/kg)	Stage 5 Chronic Kidney Disease (L/kg)	Change From Normal (%)
Increased			
Amikacin	0.20	0.29	45
Cefazolin	0.13	0.17	31
Cefoxitin	0.16	0.26	63
Ceftriaxone	0.28	0.48	71
Cefuroxime	0.20	0.26	30
Dicloxacillin	0.08	0.18	125
Erythromycin	0.57	1.09	91
Furosemide	0.11	0.18	64
Gentamicin	0.20	0.32	60
Isoniazid	0.6	0.8	33
Minoxidil	2.6	4.9	88
Naproxen	0.12	0.17	42
Phenytoin	0.64	1.4	119
Trimethoprim	1.36	1.83	35
Vancomycin	0.64	0.85	33
Decreased			
Atenolol	1.2	0.9	-25
Chloramphenicol	0.87	0.60	-31
Ciprofloxacin	2.5	1.95	-22
Digoxin	7.3	4.0	-45
Ethambutol	3.7	1.6	-57
Methicillin	0.45	0.3	-33
Metoprolol	5.6	1.0	-82
Pindolol	2.1	1.1	-48
Propranolol	4.4	3.6	-18

Data from Matzke GR, Nolin TN. Drug dosing in renal disease. In: Bombardieri A, Gilbert S, Perazella M, et al., eds. *National Kidney Foundation Primer on Kidney Diseases*. 7th ed. Philadelphia: Elsevier; 2017; Heintz BH, Matzke GR, Dager WE. Antimicrobial dosing concepts and recommendations for critically ill adult patients receiving continuous renal replacement therapy or intermittent hemodialysis. *Pharmacotherapy* 2009;29:562-577; Thummel KE, Shen DD, Isoherranen N. Appendix II. Design and optimization of dosage regimens: pharmacokinetic data. In: Brunton LL, Chabner BA, Knollmann BC, eds. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 12th ed. New York: McGraw-Hill; 2011; Murphy JE. *Clinical Pharmacokinetics Pocket Reference*. 5th ed. Bethesda, MD: American Society of Health-System Pharmacists; 2015; Verbeeck RK, Musuamba FT. Pharmacokinetics and dosage adjustment in patients with renal dysfunction. *Eur J Clin Pharmacol*. 2009;65:757-773; and Olyaei AJ, Steffl JL. A quantitative approach to drug dosing in chronic kidney disease. *Blood Purif*. 2011;31:138-145.

with the level of azotemia and may be corrected with dialysis.^{5,7,55} Binding affinity is influenced by changes in the structural orientation of the albumin molecule, as well as the accumulation of endogenous inhibitors of protein binding that compete with drugs for their binding sites.⁵⁰

Toxicity can occur if the total plasma concentration of these drugs is pushed beyond the therapeutic range. These effects are profound for some commonly used drugs, such

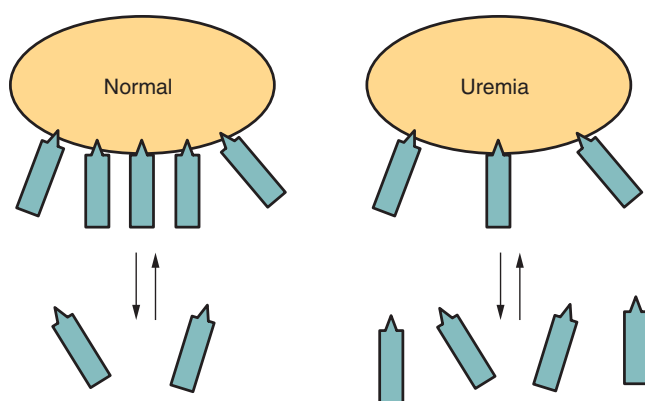


Fig. 56.2 Decreased protein binding in uremia. Displacement of the drug from its binding site by an accumulation of undefined uremic toxins or a uremia-induced conformational change in the binding-site geometry results in more free drugs in the plasma.

as seen with ceftriaxone, furosemide, metolazone, warfarin, diazepam, valproic acid, and phenytoin. Therefore when feasible, unbound plasma concentrations should be measured to guide therapy with drugs such as valproic acid and phenytoin.⁵⁵ Predicting the clinical consequences of altered protein binding is difficult. Although decreased binding results in more unbound drugs being available at the site of drug action or toxicity, the distribution volume is increased, resulting in lower plasma concentrations after a given dose. More unbound drugs are available for metabolism and excretion, which increases the clearance and decreases the half-life of the drug in the body. Drugs with decreased protein binding in patients on dialysis are listed in Table 56.2.

METABOLISM

The disposition of drugs metabolized by the liver may be altered by changes in plasma protein binding. The systemic clearance of drugs with a high hepatic extraction ratio is not thought to be impacted by the effects of CKD in contrast to that of drugs with a low extraction ratio.⁵⁶ The systemic clearance of a highly protein-bound drug with a low hepatic extraction ratio depends on the simultaneous effects of kidney function on protein binding and intrinsic metabolic drug clearance. Due to uremic toxins, protein binding of drugs will decrease in AKI and CKD, and hepatic metabolism will be inhibited. The effects of advanced CKD on protein binding with increased clearance and the effects on intrinsic metabolism with decreased drug clearance both might offset each other in terms of total systemic clearance.

Many active or toxic metabolites depend on the kidneys for their removal from the body. The accumulation of these partially active metabolites in patients with impaired kidney function (AKI and CKD) can explain, in part, the high incidence of adverse drug reactions in this patient population. For example, although the liver usually rapidly metabolizes morphine, it is categorized as excreted mainly in the urine because of the excretion of its active metabolites, morphine-3-glucuronide (M3G) and morphine-6-glucuronide (M6G). These metabolites readily cross the blood-brain barrier and bind to opiate receptors, exerting strong analgesic effects and potential adverse

Table 56.2 Unbound Fraction of Selected Drugs in Patients With Normal Kidney Function and End-Stage Kidney Disease (ESKD)

Drug	Normal Patient	ESKD Patient	Change From Normal (%)
Acidic Drugs			
Abecarnil	4	15	275
Azlocillin	62.5	75	20
Cefazolin	16	29	81
Cefoxitin	27	59	119
Ceftriaxone	10	20	100
Clofibrate	3	9	200
Dicloxacillin	3	9	200
Diflunisal	12	44	267
Doxycycline	12	28	133
Furosemide	4	6	50
Methotrexate	57.2	63.8	12
Metolazone	5	10	100
Moxalactam	48	64	33
Pentobarbital	34	41	21
Phenytoin	10	21.5	115
Salicylate	8	20	150
Sulfamethoxazole	34	58	71
Valproic acid	8	23	188
Warfarin	1	2	100
Basic Drugs			
Decreased			
Bepidil	0.3	0.1	−67
Clonidine	55.6	47.6	−14
Disopyramide	32	28	−13
Propafenone	3.4	2.4	−29
Increased			
Amphotericin B	3.5	4.1	17
Chloramphenicol	45	64	42
Clonazepam	13.9	16	15
Diazepam	2	8	300
Fluoxetine	5.5	6.5	18
Ketoconazole	1	1.5	50
Prazosin	6	10.1	68
Rosiglitazone	0.16	0.22	38
Triamterene	19	43	126

Data from Matzke GR, Nolin TN. Drug dosing in renal disease. In: Bombardieri A, Gilbert S, Perazella M, et al., eds. *National Kidney Foundation Primer on Kidney Diseases*. 7th ed. Philadelphia: Elsevier; 2017; Heintz BH, Matzke GR, Dager WE. Antimicrobial dosing concepts and recommendations for critically ill adult patients receiving continuous renal replacement therapy or intermittent hemodialysis. *Pharmacotherapy* 2009;29:562–577; Thummel KE, Shen DD, Isoherranen N. Appendix II. Design and optimization of dosage regimens: pharmacokinetic data. In: Brunton LL, Chabner BA, Knollmann BC, eds. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 12th ed. New York: McGraw-Hill; 2011; Murphy JE. *Clinical Pharmacokinetics Pocket Reference*. 5th ed. Bethesda, MD: American Society of Health-System Pharmacists; 2015; and Verbeeck RK, Musuamba FT. Pharmacokinetics and dosage adjustment in patients with renal dysfunction. *Eur J Clin Pharmacol*. 2009;65:757–773.

Table 56.3 Drugs With Pharmacologically Active Metabolites That May Affect Efficacy or Toxicity in Patients With Severe Chronic Kidney Disease

Parent Drug	Metabolite	Pharmacologic Activity of Metabolites
Acetaminophen	<i>N</i> -Acetyl- <i>p</i> -benzoquinoneimine	Responsible for hepatotoxicity
Allopurinol	Oxipurinol	Metabolite primarily responsible for suppression of xanthine oxidase
Azathioprine	Mercaptopurine	All immunosuppressive activity resides in the metabolite
Cefotaxime	Desacetyl cefotaxime	Similar antimicrobial spectrum, but 10% to 25% as potent
Chlorpropamide	2-Hydroxychlorpropamide	Similar in vitro insulin-releasing activity
Clofibrate	Chlorophenoxyisobutyric acid	Primarily responsible for hypolipidemic effect and direct muscle toxicity
Codeine	Morphine-6-glucuronide	Possibly more active than parent compound; may contribute to prolonged narcotic effect in kidney failure patients
Imipramine	Desmethylinipramine	Similar antidepressant activity
Ketoprofen	Ketoprofen glucuronide	Accumulation of acyl glucuronide may worsen toxic effects (GI disturbances, impairment of kidney function)
Meperidine	Normeperidine	Less analgesic activity than parent, but more central nervous system stimulatory effects, epileptogenic
Morphine	Morphine-6-glucuronide	Possibly more active than parent compound; may contribute to prolonged narcotic effect in ESKD
Mycophenolic acid	Mycophenolic acid glucuronide	Lacks pharmacologic activity but may be associated with dose-limiting (GI) side effects
Procainamide	<i>N</i> -Acetyl procainamide	Distinct antiarrhythmic activity; mechanism different from that of parent compound
Sulfonamides	Acetylated metabolites	Devoid of antibacterial activity; elevated concentrations associated with increased toxicity
Theophylline	1,3-Dimethyl uric acid	Cardiotoxicity has been demonstrated
Zidovudine	Zidovudine triphosphate	Primarily responsible for antiretroviral activity

ESKD, End-stage kidney disease; GI, gastrointestinal.

Data from Matzke GR, Nolin TN. Drug dosing in renal disease. In: Bombardieri A, Gilbert S, Perazella M, et al., eds. *National Kidney Foundation Primer on Kidney Diseases*. 7th ed. Philadelphia: Elsevier; 2017; Thummel KE, Shen DD, Isoherranen N. Appendix II. Design and optimization of dosage regimens: pharmacokinetic data. In: Brunton LL, Chabner BA, Knollmann BC, eds. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 12th ed. New York: McGraw-Hill; 2011; Naud J, Nolin TD, Leblond FA, et al. Current understanding of drug disposition in kidney disease. *J Clin Pharmacol*. 2012;52:10S–22S; and Battistella M, Nolin TD. Drug Therapy Individualization for Patients with Chronic Kidney Disease. In: Joseph T. DiPiro, Gary C. Yee, L. Michael Posey, et al, eds. *Pharmacotherapy: A Pathophysiologic Approach*. 11th ed. New York: McGraw-Hill; 2020.

effects. In patients with CKD, morphine's active metabolites accumulate, making prolonged narcosis and respiratory depression more likely.^{57,58} Similarly, the biotransformation of meperidine results in the production of normeperidine. In patients with impaired kidney function, repeated doses of meperidine may result in the accumulation of this potentially toxic metabolite, with resultant seizures.⁵⁹ Another useful example is the case of mycophenolic acid. The metabolite mycophenolic acid glucuronide accumulates in AKI and CKD patients, and though it lacks pharmacologic activity, it may be associated with gastrointestinal (GI) side effects. Table 56.3 highlights important drugs that form active or toxic metabolites in CKD and have been associated with adverse events.

ALTERATIONS OF CYTOCHROME P450 ENZYME ACTIVITY

A decrease in the kidney clearance of drugs in patients with CKD is well recognized. However, there is now preclinical and emerging clinical evidence suggesting that advanced CKD (stages 4 and 5) may lead to reductions in the nonkidney clearance of many medications, as a result of alterations in the activities of uptake and efflux transporters, as well as CYP enzymes, in the liver and other organs (Table 56.4).^{60–62}

The effect(s) of AKI and CKD on nonkidney drug clearance appear to depend on whether the reduction in kidney function is acute or chronic in nature and, based on evidence, on the dialysis modality (i.e., HD or peritoneal).⁶³

Preservation of nonkidney metabolic clearance has been observed early in the course of AKI,^{64–66} and thus drug dosing schemes extrapolated from those with stable CKD may therefore result in ineffectively low drug concentrations. Furthermore, failure to appreciate that change in serum creatinine (sCr) levels is a delayed marker of changes in GFR in AKI may lead to overestimation of GFR early in the disease course or underestimation of GFR during recovery, with respective dosing errors. The first reports of nonkidney clearance of drugs being differentially affected by AKI and CKD came from the observation that the residual nonkidney clearances of vancomycin, ceftriaxone, meropenem, and imipenem were higher in patients with AKI compared with patients with CKD, who had comparable levels of GFR.^{64–67}

Most of the direct evidence on drug metabolism in the presence of AKI has been derived from investigations in animal models. AKI is a heterogeneous insult that is often part of multisystem failure of cellular metabolism, which can have variable consequences.^{68–71} CYP enzymes are affected by AKI, and the extent of these effects has been

Table 56.4 Major Pathways of Nonkidney Drug Clearance (Cl_{NR})

Cl_{NR} pathway	Selected Substrates
Oxidative Enzymes	
CYP1A2	Polycyclic aromatic hydrocarbons, caffeine, imipramine, theophylline
CYP2A6	Coumarin
CYP2B6	Nicotine, bupropion
CYP2C8	Retinoids, paclitaxel, repaglinide
CYP2C9	Celecoxib, diclofenac, flurbiprofen, indomethacin, ibuprofen, losartan, phenytoin, tolbutamide, S-warfarin
CYP2C19	Diazepam, S-mephenytoin, omeprazole
CYP2D6	Codeine, debrisoquine, desipramine, dextromethorphan, fluoxetine, paroxetine, duloxetine, nortriptyline, haloperidol, metoprolol, propranolol
CYP2E1	Ethanol, acetaminophen, chlorzoxazone, nitrosamines
CYP3A4/5	Alprazolam, midazolam, cyclosporine, tacrolimus, nifedipine, felodipine, diltiazem, verapamil, fluconazole, ketoconazole, itraconazole, erythromycin, lovastatin, simvastatin, cisapride, terfenadine
Conjugative Enzymes	
UGT	Acetaminophen, morphine, lorazepam, oxazepam, naproxen, ketoprofen, irinotecan, bilirubin
NAT	Dapsone, hydralazine, isoniazid, procainamide
Transporters	
OATP1A2	Bile salts, statins, fexofenadine, methotrexate, digoxin, levofloxacin
OATP1B1	Bile salts, statins, fexofenadine, repaglinide, valsartan, olmesartan, irinotecan, bosentan
OATP1B3	Bile salts, statins, fexofenadine, telmisartan, valsartan, olmesartan, digoxin
OATP2B1	Statins, fexofenadine, glyburide
PGP	Digoxin, fexofenadine, loperamide, irinotecan, doxorubicin, vinblastine, paclitaxel, erythromycin
MRP2	Methotrexate, etoposide, mitoxantrone, valsartan, olmesartan
MRP3	Methotrexate, fexofenadine

Data from Naud J, Nolin TD, Leblond FA, et al. Current understanding of drug disposition in kidney disease. *J Clin Pharmacol*. 2012;52:10S–22S; Yeung CK, Shen DD, Thummel KE, et al. Effects of chronic kidney disease and uremia on hepatic drug metabolism and transport. *Kidney Int*. 2014;85:522–528; Kagaya H, Niioka T, Saito M, et al. Effect of hepatic drug transporter polymorphisms on the pharmacokinetics of mycophenolic acid in patients with severe renal dysfunction before renal transplantation. *Xenobiotica* 2017;47:916–922; Thomson BK, Nolin TD, Velenosi TJ, et al. Effect of CKD and dialysis modality on exposure to drugs cleared by nonrenal mechanisms. *Am J Kidney Dis*. 2015;65:574–582; Joy MS, Frye RF, Nolin TD, et al. In vivo alterations in drug metabolism and transport pathways in patients with chronic kidney diseases. *Pharmacotherapy* 2014;34:114–122; Lee W, Kim RB. Transporters and renal drug elimination. *Annu Rev Pharmacol Toxicol*. 2004;44:137–166; Masereeuw R, Russel FGM. Therapeutic implications of renal anionic drug transporters. *Pharmacol Ther*. 2010;126:200–216; and Hsueh CH, Yoshida K, Zhao P, et al. Identification and quantitative assessment of uremic solutes as inhibitors of renal organic anion transporters, OAT1 and OAT3. *Mol Pharm*. 2016;13:3130–3140.

observed to depend on the mechanism of experimental AKI and potential confounders (hypoxia, decreased protein synthesis, competitive inhibition from concomitant medications, and decreased hepatic perfusion).⁶⁷

In humans with CKD, the activities of CYPs appear to be relatively unchanged.^{60,72} It has been reported that CYP3A4 activity is reduced,⁶⁰ but studies have indicated that organic anion transporting polypeptide uptake activity is decreased.⁷² Thus the perceived changes in CYP3A4 activity were likely due to altered transporter activity, not to an alteration in CYP activity. The reduction of nonkidney clearance of several drugs that exhibit overlapping CYP and transporter substrate specificity in patients with stage 4 or 5 CKD supports this premise.^{60–62} These studies must be interpreted with caution, however, because concurrent drug intake, age, smoking status, and alcohol intake, which may independently affect drug metabolism, were often not taken into consideration. Furthermore, pharmacogenetic variations in drug-metabolizing enzymes in the individual irrespective of AKI or CKD must also be considered.

KIDNEY EXCRETION

Kidney clearance (Cl_K) of a drug is the composite of the GFR, the fractions of tubular secretion, metabolism, and reabsorption, where f_u is the unbound fraction of drug (i.e., not bound to plasma proteins). The correlation between secretory drug clearance and GFR is linear, and the intercept is not different from zero.⁷³

$$Cl_K = GFR \cdot (f_u + f_{\text{secretion}} + f_{\text{metabolism}} - f_{\text{reabsorption}})$$

Drug elimination by filtration occurs by a pressure gradient, whereas tubular secretion and reabsorption are bidirectional active processes that involve carrier-mediated kidney transport systems.^{60,74,75}

Kidney transport systems have been broadly classified on the basis of substrate selectivity into anionic and cationic kidney transport systems, which are responsible for the transport of a number of organic acidic and basic drugs, respectively.^{51,60} Several drugs are actively secreted by one or more of these transporter families, including organic cationic (e.g., famotidine, trimethoprim, and dopamine), organic anionic (e.g., ampicillin, cefazolin, and furosemide),

nucleoside (e.g., zidovudine), and P-glycoprotein (e.g., digoxin, *Vinca* alkaloids, and steroids) transporters.^{7,74,75} Alterations in filtration, secretion, or reabsorption secondary to CKD may have a dramatic effect on drug disposition.⁷⁶ For drugs that are primarily filtered, a reduction in GFR will result in a proportional decrease in kidney drug clearance. However, evidence indicates that the clearance of cationic secreted drugs correlates poorly with measured GFR⁷⁷ and that uremic solutes inhibit OAT1 and OAT3 transporters and contribute to the decline in kidney drug clearance in CKD patients.⁷⁶ OAT1 and 3 are multispecific exchangers or antiporters that transport predominantly anionic substrates against a concentration gradient from the blood into proximal tubule cells for subsequent elimination into the urine. The clinical impact of these findings limits but does not nullify the utility of eGFR for dose adjustment in patients with CKD. For further discussion on renal transport of organic solutes, see [Chapter 8](#).

PHARMACOGENOMICS

Over the past 2 decades, genome-wide analyses have identified genetic variants that are associated with either a relative increase or decrease in risk of several diseases.^{78,79} Most confer a low relative risk, are common alleles, and have low discriminatory and predictive values.^{80,81} Thus how CKD patients respond to medications is a consequence of alterations in pharmacokinetics and pharmacodynamics, as well as pharmacogenomics. For example, the activity of different enzymes involved in warfarin and vitamin K metabolism has varying effects on therapeutic anticoagulation. Genotyping data on two genes, the warfarin metabolic enzyme CYP2C9 and warfarin target enzyme vitamin K epoxide reductase complex 1 (VKORC1), confirmed that each can influence warfarin maintenance dose.⁸² Two genome-wide association studies also found a weak, but significant, effect of CYP4F2 on warfarin dosing.^{83,84} The presence of *CYP2C9**2 or *CYP2C9**3 variant alleles, which results in decreased metabolic enzyme activity, is associated with a significant decrease in the mean warfarin dose,⁸⁵ whereas *VKORC1* single-nucleotide polymorphisms identify *VKORC1* haplotypes accounting for a large fraction of the interindividual variation in warfarin dose.⁸⁶ The combination of both *VKORC1* and *CYP2C9* polymorphisms is associated with severe overanticoagulation.⁸⁷ In contrast to these common but weak-effect genetic variants, some DNA sequence variants are rare but have a strong effect; one example would be azathioprine. Polymorphisms in the gene that encodes thiopurine S-methyltransferase (*TPMT*) lead to reduced activity of this metabolizing enzyme, and patients with low activity (10% prevalence) or absent activity (0.3% prevalence) are at higher risk of bone marrow depression because of drug accumulation.^{88,89} In many places measurement of *TPMT* activity is routine before initiation of azathioprine, but if measurement or genotyping of *TPMT* activity in advance is not possible, leucocyte count should be closely monitored within the first 2 weeks. The variability in how patients respond to drug treatments is clearly a consequence of alterations in pharmacokinetics and pharmacodynamics, as well as differences in their genotypes

and/or phenotypes.^{79,90-95} In general, DNA sequence code variants affect prodrug activation (i.e., clopidogrel), drug metabolism and degradation (i.e., warfarin), or predisposition to adverse drug effects (i.e., azathioprine).

Genotyping information is becoming more widely available than phenotyping data, and this will allow for a more individualized approach to pharmacotherapy. Genotypic characterization now serves as the basis for dosing recommendations for many drugs, as highlighted by >120 U.S. Food and Drug Administration (FDA)-approved drugs having pharmacogenomic information in their labeling, including fluoropyrimidines, codeine, selective serotonin reuptake inhibitors, tricyclic antidepressants, β -blockers, opiates, neuroleptics, antiarrhythmic agents, and statins.⁹⁶⁻¹⁰⁰

The future of pharmacogenetics will lie in treatment models in which patient characteristics such as CKD or AKI are combined with polymorphism information regarding relevant genes. At present, data are insufficient to warrant genomic testing in persons with CKD to guide drug therapy. Future work will focus on the use of both pharmacokinetic and pharmacogenomic testing to improve drug dosing for CKD patients

PHARMACODYNAMICS

The fundamental concept of pharmacodynamics is described by the Hill equation.^{101,102} The concentration (C) of a drug is the primary factor necessitating alterations in dosage regimens to achieve the desired pharmacodynamics targets. The actual effect (E) of a drug is a function of the maximum effect ($E_{\max}$) and the concentration producing the half-maximum effect (CE_{50}). The Hill coefficient (H) is a measure of the sigmoidicity of the effect–concentration correlation:

$$E = \frac{E_{\max}}{1 + \left(\frac{CE_{50}}{C}\right)^H}$$

This model has been extensively used to optimize the effects of many antimicrobial agents.¹⁰³ Pharmacodynamic principles apply to guide the dosing of medications in patients with CKD, as they do in people with normal kidney function. In the patient with CKD, the concentration–time profile of many drugs is altered, so the dosage regimen predicted will likely be different from the normal regimen. This is because of the prolonged elimination half-life, which results in an increased AUC. Only rarely has there been evidence of an alteration in the pharmacodynamic concentration–effect relation in patients with AKI or CKD; pharmacokinetic changes more than pharmacodynamic changes contribute to the need for a modified dosing regimen with kidney dysfunction.

From the Hill equation, the threshold concentration, which produces 5% of the maximum effect, and the ceiling concentration, which is associated with 95% of the maximum effect, can be derived. The higher the Hill coefficient, the higher the threshold concentration and the narrower the range of lower and upper target concentrations; this is because the ceiling concentration

comes down close to the concentration producing the half-maximum effect (Fig. 56.3):

$$CE_{05} = 19^{-\frac{1}{H}} \bullet CE_{50}$$

$$CE_{95} = 19^{\frac{1}{H}} \bullet CE_{50}$$

The difference between the ceiling and threshold concentrations can be measured by multiples of the respective elimination half-life. The ceiling concentration is the upper limit of the targeted peak concentration ($C_{peak} < CE_{95}$), whereas the threshold concentration marks the lower limit of effective trough concentration ($C_{trough} > CE_{05}$). For a drug with a short half-life ($t_{1/2}$) and a high Hill coefficient, the therapeutic range of target concentrations can be small (see Fig. 56.3):

$$CE_{05} = CE_{95} \bullet \exp \left[-\frac{\ln(2)}{T_{1/2}} \bullet t \right]$$

$$t_{\text{ceiling-threshold}} = T_{1/2} \bullet \frac{2}{H} \bullet \frac{\ln(19)}{\ln(2)}$$

$$t_{\text{ceiling-threshold}} = T_{1/2} \bullet \frac{8.5}{H}$$

For example, for the β -lactam ceftazidime, with a short half-life of 2.1 hours in patients with normal kidney function but with a high Hill coefficient of 3.7,¹⁰¹ the peak to trough or ceiling to threshold time of 5 hours indicates that ceftazidime should be given at least every 6 hours to maximize efficacy. This contrasts with the once-daily dosing of aminoglycosides. In agreement with the postulated postantibiotic effect (a period of time after complete removal of an antibiotic during which there is no growth of the target organism), the maximum peak to trough time is estimated as 13 hours for gentamicin, with a half-life of 2 hours but a Hill coefficient of 1.3.¹⁰¹

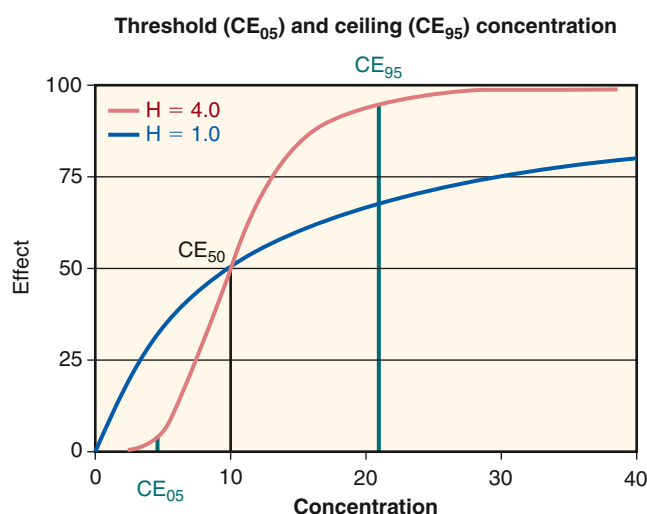


Fig. 56.3 Threshold concentration, CE_{05} , producing 5% of the maximum effect, and ceiling concentration, CE_{95} , producing 95% of the maximum effect. With a Hill coefficient of $H = 1.0$, $CE_{05} = 0.5$ and $CE_{95} = 190$, whereas for $H = 4.0$, the threshold is higher, with $CE_{05} = 6.0$, but the ceiling is much less, with $CE_{95} = 21$ mg/L.

The most important progress in antimicrobial dosing has been achieved with the differentiation of drugs with time-dependent actions from drugs with concentration-dependent actions.^{104,105} Specific examples are the β -lactam antibiotics and antiviral drugs with a known time-dependent effect, whereas aminoglycosides and quinolones have a concentration-dependent activity. The threshold and ceiling concentrations are specific functions of the concentration producing the half-maximum effect and the Hill coefficient. Both can be used to explain the findings that antimicrobial drugs with a time-dependent effect have a significantly higher Hill coefficient than those with a concentration-dependent action.¹⁰¹ A high Hill coefficient is associated with a high threshold concentration but, simultaneously, with a relatively low ceiling concentration. Thus it makes no sense to increase the dose of time-dependent antimicrobial drugs above the ceiling concentration. By contrast, a low Hill coefficient is associated with a high ceiling concentration and low threshold concentration. Consequently, the effect of concentration-dependent antimicrobial drugs may be increased by giving a high single dose, as the concentration-dependent effect does not decrease when the administration interval is extended, as proposed for aminoglycosides.¹⁰⁶ Practically, antimicrobial drugs with a time-dependent action must be administered more frequently, whereas antimicrobial drugs with a concentration-dependent action should be given at higher maintenance doses to increase efficacy (Fig. 56.4).

Usual measures of the antimicrobial effect, such as the minimal inhibitory concentrations (MICs), AUC over MIC, time over MIC ($T > MIC$), or peak C_{max} over MIC, can be unified to the following concept. The target concentration should not be less than the threshold concentration for time-dependent effects, but it could be as high as the ceiling concentration for concentration-dependent effects. A close correlation of the MIC and concentration producing the half-maximum effect has been postulated.¹⁰¹ It was obvious,

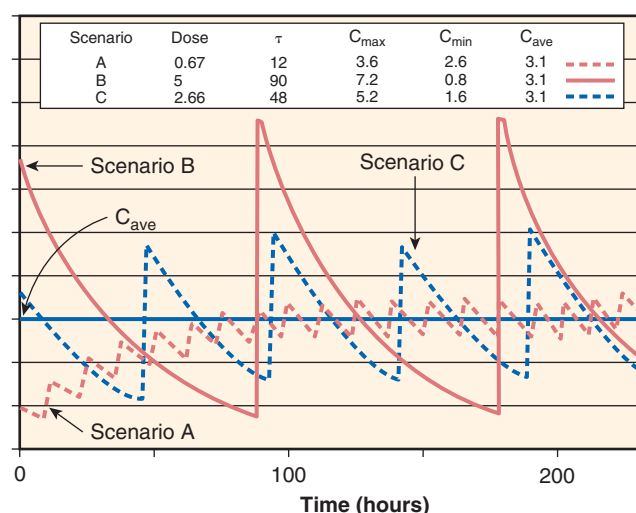


Fig. 56.4 Although the average steady-state concentrations (C_{ave}) are identical regardless of which dosage adjustment strategy one decides to use, the concentration-time profile will be markedly different if one changes the dose and maintains the dosing interval (τ) constant (Scenario A), versus changing the dosing interval and maintaining the dose constant (Scenario B), or changing both (Scenario C).

however, for concentration-dependent antimicrobial action, the MIC could fall considerably below the concentration producing the half-maximum effect ($MIC \ll CE_{50}$). Consequently, it might be more reasonable to compare the bacteriologic MIC with the pharmacodynamic parameter of a threshold CE_{05} concentration:

$$C_{\text{threshold}} = CE_{05} = MIC$$

From the Hill coefficient, one can postulate that the time-dependent action and concentration-dependent action are only the extreme positions of a continuum. Every drug can be considered concentration dependent and time dependent. To overcome resistance, a higher dose might be necessary, because relative resistance can be seen in cases in which a high concentration is required to produce the half-maximum effect. The potency is the inverse concentration producing the half-maximum effect:

$$\text{Potency} = \frac{1}{CE_{50}}$$

This concept distinguishes relative from absolute drug resistance. A pathogen with relative resistance can be made sensitive by increasing the dose.¹⁰⁷⁻¹⁰⁹ Thus for example, it has been recommended to treat severe infections with resistant strains by increasing the standard meropenem dose from 1000 mg three times daily, to 2000 mg three times daily,¹¹⁰ or the daptomycin dose from 4–6 mg/kg/day to >8 mg/kg/day.¹¹¹ Such large doses will need careful monitoring of side effects and adjustment for kidney function.

Clinical Relevance

Pharmacogenetic factors predicting therapeutic response and toxicity in patients with kidney disease are emerging. Changes in pharmacokinetic pathways of absorption, distribution, and elimination (metabolism, transport, and excretion) of prescribed drugs observed in patients with acute kidney injury and chronic kidney disease can confound the outcomes of personalized therapeutic interventions.

ASSESSMENT OF KIDNEY FUNCTION FOR DRUG DOSING

The standard measure of kidney function has been the GFR (for further discussion on GFR, see [Chapter 23](#)). GFR can be measured using many exogenous substances; however, the administration of exogenous substances is not practical for routine individual drug dose calculations in clinical practice because the procedures are not timely and not uniformly available. For dose adjustment, the exact GFR measurement is less important than the easily and immediately available eGFR.

CHRONIC KIDNEY DISEASE

The estimated eGFR ([Table 56.5](#)) is used in clinical practice based on the serum creatinine and/or cystatin C (CysC) concentrations given the convenience.¹¹²⁻¹¹⁸ The measured creatinine clearance (mCrCl) is the only endogenous

method to determine glomerular hyperfiltration in pregnancy as cystatin C is not reliable in pregnancy.

Estimated GFR based on different current creatinine formulas may yield different CKD stage categorizations, which would impact drug dose recommendations based on original studies that used different formulas.¹¹⁹ It is not practical to repeat all the pharmacokinetics studies with standardized creatinine-determined eGFR, and therefore it is still reasonable to use drug dosing adjustments that appear in FDA- and European Medicines Agency (EMA)-approved product labeling, but clinicians should be aware of these concerns.

Traditionally, drug dosing was based on not the measured endogenous creatinine clearance (mCrCl) but on estimated creatinine clearance (eCrCl) using the Cockcroft and Gault (CG) formula.^{5,7,18,19} The CG equation is not suitable for clinical reporting because body weight may not be available in the electronic health record and it does not apply across the spectrum of GFR. The Modification of Diet in Renal Disease (MDRD) equations^{120,121} were initially used by clinical laboratories, although they were only validated for patients with a GFR <60 mL/min. Therefore the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation was developed to allow estimation of GFR throughout the full range of the CKD.¹¹⁶ In 2021, the CKD-EPI research group developed a new CKD-EPI eGFRcr equation without the inclusion of race,¹²² which is the currently preferred formula, although it does perform variably in diverse populations.^{121,123,124,125} Serum cystatin C has been proposed as an alternative marker to estimate GFR.¹²⁶ The combined use of both serum markers, cystatin C and creatinine, permits more accurate estimation of kidney function than either of them alone.¹²⁷⁻¹²⁹

Furthermore, the new cystatin C–based European Kidney Function Consortium EKFC equation estimates the eGFR independent from sex and race has been shown to improve GFR assessment.¹³⁰

Both, the 2021-CKD-EPI (either creatinine or cystatin C alone or in combination) and the EKFC equations estimate the GFR for a standard 1.73 m² body surface area (BSA); thus for an individual patient, the BSA must be determined separately so that the GFR can be expressed in milliliters per minute (mL/min).

$$BSA = \frac{\sqrt{\text{weight (kg)} \cdot \text{height (m)}}}{60}$$

$$GFR = eGFR \cdot \frac{BSA}{1.73}$$

Adjusting drug doses based on the combined measurement of creatinine and cystatin C appears to be an effective and valid tool in the limited number of applications (mainly relating to chemotherapy and antibiotic dosing, e.g., vancomycin) for which it has been studied.¹³¹⁻¹³⁵

Some specialists argue that use of the CKD-EPI equations for drug dosing is not appropriate given that the pharmacokinetics studies were performed using eCrCl via the CG equation.^{18,117} Furthermore, many studies have highlighted discordance between drug dosing recommendations based on these equations.¹³⁹⁻¹⁴³ For now it is recommended that estimation of GFR using the most appropriate formula should be an initial step in decision

Table 56.5 Equations For Estimation of Creatinine Clearance or Glomerular Filtration Rate in Adults With Stable Renal Function^{a,b}

Reference	Equation
Cockcroft & Gault (1976)	$mCrCl = uCr \times \text{Volume}_{24 \text{ h urine}} / sCr$ Men: $CrCl = (140 - \text{age}) \times IBW / (sCr \times 72)$ Women: $CrCl \times 0.85$
Jelliffe (1973)	Men: $CrCl = 98 - [0.8 \times (\text{age} - 20)] / sCr$ Women: $CrCl \times 0.9$
CKD-EPI (2009)	$eGFR_{Cr} = 141 \times \min(sCr/\kappa, 1)^{\alpha} \times \max(sCr/\kappa, 1)^{-1.209} \times 0.993^{\text{age}} \times (1.018 \text{ if patient is female}) \times (1.159 \text{ if patient is black})$ κ is 0.7 for females and 0.9 for males. α is -0.329 for females and -0.411 for males. min is the minimum of sCr/κ or 1. max is the maximum of sCr/κ or 1.
Larsson et al. (2004)	$eGFR_{Cys} = 77.24 \times (CysC [\text{in mg/L}])^{-1.2623}$
MacDonald et al. (2006)	$\text{Log}_{10} eGFR_{Cys} = 2.222 + (-0.802 \times \sqrt{CysC \text{ in } \frac{\text{mg}}{\text{L}}}) + (0.009876 \times LM)$
CKD-EPI cystatin C Eq. (2012)	$eGFR_{cys} = 133 \times \min(sCys/0.8, 1) - 0.499 \times \max(sCys/0.8, 1) - 1.328 \times 0.996^{\text{age}} (\times 0.932 \text{ if female})$ $sCys$ is serum cystatin C. min is the minimum of $sCys/0.8$ or 1. max indicates the maximum of $sCys/0.8$ or 1.
CKD-EPI creatinine-cystatin C Eq. (2012)	$eGFR_{Cr-Cys} = 135 \times \min(sCr/\kappa, 1)^{\alpha} \times \max(sCr/\kappa, 1) - 0.601 \times \min(sCys/0.8, 1) - 0.375 \times \max(sCys/0.8, 1) - 0.711 \times 0.995^{\text{age}} (\times 0.969 \text{ if female}) (\times 1.08 \text{ if black})$ κ is 0.7 for females and 0.9 for males. α is -0.248 for females and -0.207 for males. min indicates the minimum of sCr/κ or 1. max indicates the maximum of sCr/κ or 1.
EKFC equation Pottel (2023)	$EKFC-eGFR = 107.3 / [\text{Biomarker}/Q]^{\alpha} \times [0.990^{(\text{Age}-40)} \text{ if age } > 40 \text{ years}]$ with $\alpha = 0.322$ when biomarker/ Q is < 1 and $\alpha = 1.132$ when biomarker/ Q is ≥ 1 . For biomarker cystatin C $Q' = 0.79 \text{ mg/L}$ until age 50 years, and $Q' = 0.79 + 0.005 \times (\text{age} - 50)$ thereafter, for women, $Q' = 0.86 \text{ mg/L}$ until age 50 years, and $Q' = 0.86 + 0.005 \times (\text{age} - 50)$ thereafter for men
2021 CKD-EPI cr	$eGFR = 142 \times \min(Scr/\kappa, 1)^{\alpha} \times \max(Scr/\kappa, 1)^{-1.200} \times 0.9938^{\text{age}} \times 1.012 [\text{if female}]$ $eGFR$ (estimated glomerular filtration rate) = $\text{mL/min}/1.73 \text{ m}^2$ Scr (standardized serum creatinine) = mg/dL $\kappa = 0.7$ (females) or 0.9 (males) $\alpha = -0.241$ (females) or -0.302 (males) min = indicates the minimum of Scr/κ or 1 max = indicates the maximum of Scr/κ or 1 age = years
2021 CKD-EPI-cr-cys	$eGFR = 135 \times \min(Scr/\kappa, 1)^{\alpha} \times \max(Scr/\kappa, 1) - 0.544 \times \min(Scys/0.8, 1) - 0.323 \times \max(Scys/0.8, 1) - 0.778 \times 0.9961^{\text{age}} \times 0.963 [\text{if female}]$ $eGFR$ (estimated glomerular filtration rate) = $\text{mL/min}/1.73 \text{ m}^2$ Scr (standardized serum creatinine) = mg/dL $Scys$ (standardized serum cystatin C) = mg/L $\kappa = 0.7$ (females) or 0.9 (males) $\alpha = -0.219$ (females) or -0.144 (males) min = indicates the minimum of Scr/κ or 1 max = indicates the maximum of Scr/κ or 1 age = years

^aFor SI conversion purposes, serum or plasma creatinine is converted from $\mu\text{mol/L}$ to mg/dL by multiplying by 0.0113; conversion from creatinine clearance conventional units of mL/min to SI units of mL/s requires multiplication by 0.0167.

^bEquations compiled from references.^{251,252}

Alb, Albumin; *BUN*, blood urea nitrogen; *CKD-EPI*, Chronic Kidney Disease Epidemiology Collaboration; *CrCl*, creatinine clearance in mL/min ; *CysC*, cystatin C; *eGFR*, estimated glomerular filtration rate; *IBW*, ideal body weight (kg); *IDMS*, isotope dilution mass spectroscopy; *LM*, lean mass; *MDRD*, Modification of Diet in Renal Disease; *sCr*, serum or plasma creatinine (mg/dL).

making rather than a conclusive step in decision making. Careful consideration of the context and clinical status is necessary.

PEDIATRICS

The original equation to estimate GFR, as described by Schwartz and colleagues,¹⁴⁴ is dependent on the child's age and length:

$$GFR = (\text{length}[\text{cm}] \bullet k) / sCr (\text{in mg/dL})$$

where k is defined by age group: infant (1–52 weeks) = 0.45; child (1–13 years) = 0.55; adolescent male = 0.7; and adolescent female = 0.55. The serum creatinine level in $\mu\text{mol/L}$ can be converted to mg/dL by multiplication using 0.0113 as the conversion factor. A newer version of the Schwartz equation¹⁴⁵ was developed from a population of 349 children (age 1–19 years) with mild to moderate CKD enrolled in the Chronic Kidney Disease in Children (CKiD) study:

$$GFR = 0.41 \bullet (\text{length in cm}) / sCr (\text{in mg/dL})$$

Lee and associates¹⁴⁶ have reported that this new Schwartz equation performed better than the original Schwartz equation for patients with moderate CKD but was less accurate in patients with mild CKD. In pediatric patients, methods incorporating cystatin C have several advantages for evaluating kidney function.¹⁴⁷ The most recent eGFR equation evaluated in pediatrics includes use of cystatin C, blood urea nitrogen (BUN), serum creatinine level (in mg/dL), and demographic data derived from over 600 pediatric patients enrolled in the CKiD study¹⁴⁸:

$$\begin{aligned} \text{eGFR (mL/min/1.73 m}^2\text{)} \\ = 39.8 \cdot (\text{ht [m]}/\text{sCr})^{0.456} \cdot (1.8/\text{cystatin C})^{0.418} \\ \cdot (30/\text{BUN})^{0.079} \cdot 1.076^{\text{male}} \cdot (\text{ht [m]}/1.4)^{0.179} \end{aligned}$$

This equation had the highest R^2 value (0.863) and the highest frequency of values within 30% of iohexol-measured GFR (91.3%) when compared with seven other GFR estimating equations.

ACUTE KIDNEY INJURY

During AKI, the GFR is a moving target so a kinetic GFR has been proposed (see later), although changes in serum creatinine are most routinely used.¹⁴⁹ The mCrCl may also permit assessment of the augmented kidney clearance in the initial phase of sepsis, especially in the intensive care unit (ICU), where urine output is routinely monitored. At present, the staging of AKI is based on sequential measurement of the serum creatinine level and urine output.¹⁵⁰⁻¹⁵² Because the GFR is inferred from the serum creatinine or cystatin C, all estimates of kidney function are delayed and lag the real-time GFR. Although several methods have been proposed to estimate GFR in this patient population, none have been rigorously evaluated.¹⁵³⁻¹⁵⁶ The latest proposed method to estimate GFR in patients with AKI is the kinetic GFR (kinetGFR), which is based on age (years), weight (kg), and serum creatinine ($\mu\text{mol/L}$) and holds true for increasing and decreasing kidney function.¹⁴⁹

$$\begin{aligned} \text{kinetGFR} = \frac{[150 - \text{age (years)}] \cdot \text{weight (kg)}}{Cr_1 (\mu\text{mol/L})} \\ \cdot \left[1 - \frac{Cr_2 - Cr_1}{t_2 - t_1} \cdot \frac{24 \text{ (hours)}}{200 (\mu\text{mol/L})} \right] \end{aligned}$$

This approach is based on an estimate of the creatinine production similar to the CG equation.¹¹² The kinetic eGFR incorporates changing creatinine values over specified time intervals, as well as the actually measured serum creatinine values. It relates the increase in serum creatinine within a specified time interval to the maximum increase in creatinine level in one day. The kinetic eGFR tries to tackle the problem that there is always a delay between rapidly changing kidney function and measurable variables—namely, serum creatinine or urine output. The calculation of a patient's kinetic eGFR may permit use of the eGFR-based dose-adjustment recommendations derived from patients with CKD for patients with AKI.¹⁴⁹ Rigorous independent studies will be needed to confirm its validity and utility in clinical practice.

PATIENTS RECEIVING DIALYSIS

Some ESKF patients on dialysis and those with AKI on CKRT have residual kidney function that substantially contributes to better patient outcomes and the elimination of drugs and their metabolites.¹⁵⁷ Measuring the elimination of iohexol after an IV dose has been reported to be an accurate and safe measure of residual kidney function in patients on dialysis and can inform drug dosing.^{158,159} The eGFR derived from serum creatinine during continuous KRT represents the combined effect on drug clearance by residual kidney function and by KRT.

GOALS OF THERAPY

Despite the availability of numerous guidelines regarding drug dosing for patients with impaired kidney function, there is insufficient evidence as to which, if any, is preferred.^{6,19,51,160-164} Occasionally, recommendations derived from postmarketing studies conflict with the information in these reports, as well as the official FDA or EMA product labeling. Before 1998, there were no official guidelines regarding when and how to characterize the relationship between the pharmacokinetics and pharmacodynamics of a drug and kidney function. The FDA guidelines issued in May 1998,¹⁶⁵ the 2010 revision,¹⁶⁶ and the EMA guidelines of 2015¹⁶⁷ have provided frameworks for which drugs should be evaluated and guidance regarding study design, data analysis, interpretation of study results, and recommendations for the incorporation of data into product labeling.

The desired goal is typically the maintenance of a similar peak, trough, or average steady-state drug concentration or, for antibiotics, an optimized pharmacodynamic measure, such as the peak over MIC, the time above the MIC, or the ratio of the drug area under the AUC to the MIC, as would be optimal for persons with normal kidney function (see the “Pharmacodynamics” section above for more detail). When there is a significant relationship between drug concentration and clinical response¹⁰¹ (e.g., aminoglycosides) or toxicity⁵⁵ (e.g., phenytoin), attainment of the specific target values becomes critical. If, however, no specific pharmacokinetic or pharmacodynamic target values have been reported, a dosing goal of attaining and maintaining the same average steady-state concentration may be appropriate.

FDA-approved drug monographs and commonly used drug information sources such as American Hospital Formulary Service Drug Information, Goodman and Gilman's *The Pharmacological Basis of Therapeutics*, the *British National Formulary*, and *Drug Prescribing in Renal Failure* are excellent sources of information about a drug's pharmacokinetic characteristics, goals of therapy, and the dosing regimens to achieve the goals (i.e., adequate drug concentrations or pharmacodynamic target values, in various patient populations).¹⁶¹⁻¹⁶⁴ However, in the digital era, many clinicians rely on drug information via their smartphone or personal digital assistant (e.g., www.dosing.de). Examples of commonly used programs that answer frequent bedside drug questions, such as the proper dosing, pharmacology, adverse side effects, drug–drug interactions, and pregnancy safety, are Epocrates Rx,¹⁶⁸ Medscape Mobile,¹⁶⁹ Micromedex Drug Information,¹⁷⁰ Lexicomp,^{xx} and mobilePDR.¹⁷¹ There are more specific

applications for drug dosing in kidney dysfunction, but they require payment to access the information: KidneyCalc¹⁷² and ABX Dosage—Adjustments in Renal Failure.¹⁷³ Regardless of dosing recommendations by the various sources, the design of the optimal dosage regimen is dependent on the availability of an accurate characterization of the relationship between the pharmacokinetic parameters of the drug and kidney function and an accurate assessment of the patient's kidney function.

INDIVIDUALIZATION OF THE DRUG DOSAGE REGIMEN WITH KIDNEY DYSFUNCTION

Most dose-adjustment guidelines propose the use of a fixed dose or interval for patients with broad ranges of kidney function.^{51,160-163} The mild, moderate, and severe CKD categories vary among reference sources, so the recommended dosing may not be “optimal” for all patients whose kidney function lies within the range, especially for agents with a narrow therapeutic index.¹⁸ Narrow therapeutic index drugs are drugs where small differences in dose or blood concentration may lead to serious therapeutic failures and/or adverse drug reactions that are life-threatening or result in persistent or significant disability or incapacity.

The approach to developing drug dose-adjustment recommendations for the patient with CKD is predicated on attainment of the desired exposure goal at steady state. To achieve the desired goal in a timely fashion, a stepwise approach that includes multiple considerations (Table 56.6) for each individual drug should be considered.¹³⁵ The following considerations may help guide individualization of therapy.

The initial/starting or loading dose (D_{start}), which in many patients with AKI will be larger than the typical maintenance dose, should be calculated to achieve the desired C_{max} therapeutic drug concentration. A D_{start} should be used for most patients with stage 4 or 5 CKD to achieve the desired steady-state concentration rapidly and in which the V_D of a drug is significantly increased in extracellular volume-expanded patients with AKI and/or CKD relative to those with normal kidney function. If a dependence of V_D on CrCl has been characterized, then the V_D should be estimated from that relationship. If no D_{start} is prescribed, four to five half-lives of the drug will pass before the desired steady-state plasma concentration is achieved; however, doing so may contribute to therapeutic failure. The size of the D_{start} given affects the height of the steady-state plasma concentration and how rapidly plasma concentrations are achieved. A D_{start} equivalent to the dose given to a patient with normal kidney function should be given to patients with impaired kidney function if the drug's half-life is especially long and if the physical examination suggests normal ECF volume. If the patient has marked volume expansion or evidence indicates that the V_D of the drug is larger in patients with CKD, then a higher dose can be calculated from the following expression:

$$D_{\text{start}} = V_D \cdot C_{\text{max}} \cdot \text{actual weight} / \text{IBW},$$

where V_D is the drug's volume of distribution (in liters per kilogram of IBW in those with CKD), IBW is the patient's ideal body weight (in kilograms), and C_{max} is the desired steady-state maximum plasma drug concentration.

The primary reference for information regarding the maintenance dose for patients with CKD should be the FDA

Table 56.6 Stepwise Approach to Adjust Drug Dosage Regimens for Patients With Impaired Kidney Function

Step	Process	Assessment
1	Obtain history and relevant demographic and clinical information	Record demographic information, obtain medical history (including history of renal disease), and record current laboratory information (e.g., serum creatinine)
2	Estimate creatinine clearance and/or determine eGFR	Use Cockcroft–Gault equation to estimate creatinine clearance, or calculate creatinine clearance from timed urine collection and/or determine eGFR using 2011 CKD EPIcr or CKD EPIcr-cys
3	Review current medications	Identify drugs for which individualization of the treatment regimen will be necessary
4	Calculate individualized treatment regimen	Determine treatment goals (see text); calculate dosage regimen based on pharmacokinetic characteristics of the drug and patient's renal function.
5	Monitor	Monitor parameters of drug response and toxicity; monitor drug levels if available or applicable.
6	Revise regimen	Adjust regimen based on drug response or change in patient status (including kidney function), as warranted.

Adapted from Battistella M, Nolin TD. Drug Therapy Individualization for Patients with Chronic Kidney Disease. In: Joseph T. DiPiro, Gary C. Yee, L. Michael Posey, et al., eds. *Pharmacotherapy: A Pathophysiologic Approach*. 11th ed. New York: McGraw-Hill; 2020.

and/or EMA official product labeling. If no official drug dosing guidance is available, one may need to search the literature to find a recommendation strategy derived from nonregulatory or postmarketing clinical investigations. If no such resource is found, one can consult online or published tertiary references that have developed dosing recommendations based on the Dettli or Tozer method, initially published in 1974.^{21,22} They used similar foundational pharmacokinetic characteristics and approaches to calculate the maintenance dose for a patient with a given eCrCl. In essence, either the dose (D) should be reduced or the interval (τ) extended. When the dose is reduced, the C_{max} will be lower and the trough concentrations will be higher than those observed in persons with normal kidney function. When the administration interval is extended, the peak and trough concentrations are kept constant but the dosing frequency decreases (see Fig. 56.4).

To maintain the normal dose interval in patients with impaired kidney function, the amount of each dose after the D_{start} can be estimated from the following equation:

$$D_f = D_n \cdot Q$$

where D_f is the dose for the patient with impaired kidney function to be given at the normal dosing interval, D_n is the normal dose, and Q is the dosage adjustment factor. The dosage adjustment factor (Q) can be calculated as:

$$Q = 1 - (FE \cdot [1 - KF]),$$

where FE is the fraction of the drug eliminated unchanged in the kidney in a patient with normal kidney function and KF is the ratio of the patient's CrCl or eGFR to the assumed normal value of 120 mL/min (equivalent to 2.00 mL/sec). Thus for a drug that is 85% eliminated unchanged by the kidneys, the Q factor in a patient who has a CrCl of 10 mL/min (0.17 mL/sec) would be as follows:

$$\begin{aligned} Q &= 1 - \left(0.85 \cdot \left[1 - \frac{10}{120} \right] \right) \\ &= 1 - (0.85 \cdot [0.92]) \\ &= 1 - 0.78 \\ &= 0.22 \end{aligned}$$

If one desires to give the same maintenance dose, a factor that may be required because of the limited availability of alternative formulations, the dosing interval at which the normal dose should be administered can be calculated as follows:

$$\tau_f = \tau_n / Q$$

The decision to extend the dosing interval beyond a 24-hour period should be based on the need to maintain therapeutic peak or trough levels. The dosing interval may be prolonged if the peak level is most important. For

patients receiving HD three times per week, prolonging the dose interval is a convenient method to modify the drug dosage regimen, especially if a drug is intravenously administered. This method is particularly useful for drugs with a long plasma half-life. In general, drugs removed by dialysis given once daily should be given after the dialysis treatment, with aminoglycosides or anticancer drugs (Table 56.7) being a controversial exception.¹⁷⁴⁻¹⁷⁷

A third alternative that is especially helpful when the calculated dose or dosing interval is impractical is to select the administration interval according to the target trough concentration while the peak is kept constant:

$$\begin{aligned} \tau_{\text{target}} &= \left(T_{1/2} / 0.693 \right) \cdot \ln(C_{\text{peak}} / C_{\text{trough-target}}) \\ D &= D_{\text{start}} \cdot (1 - C_{\text{trough-target}} / C_{\text{peak}}) \end{aligned}$$

Alternatively, one can calculate the adjusted dose (D_p) to be given at the predetermined practical dosage interval (e.g., 12, 18, 24, 36, and 48 hours). These approaches, which use a combination of dose reduction and interval prolongation methods, are often the most clinically practical. When in doubt, clinicians should consult an experienced renal pharmacist, preferably one with extensive experience in evaluating patients with CKD and altered body composition (e.g., fluid overload).

MEASUREMENT OF THERAPEUTIC DRUG LEVELS

Measuring drug concentrations is recommended in AKI and CKD as one way to optimize therapeutic regimens

Table 56.7 Drugs Best Administered Before Hemodialysis

Drug Class	Examples	Drug Fraction Removed by One Dialysis Session	Reference
Anticancer	Carboplatin	20% dose post-HD; 84% dose before HD	Chatelut et al.209; Kamata et al.210; Yoshida et al.211; Oguri et al.212
	Cisplatin	85%	Watanabe et al.213
	Oxaliplatin	65%	Katsumata et al.214
	Cyclophosphamide	22% (M % unknown)	Haubitz et al.215
	Ifosfamide	70%–87% (M, 72%–77%)	Carlson et al.216
	Capecitabine	FBAL 50%	Walko and Lindley ²¹⁷
	Gemcitabine	dFdU 50%	Koolen et al.218
	Methotrexate	(M, 36%)	Garlich and Goldfarb ²¹⁹
	Cytosine arabinoside	39% (M, 52%–63%)	Radeski et al.220
	Topotecan	50%	Herrington et al.221
	Bleomycin	30%–60%	Kamidono et al.223
	Lenalidomide	30%	João et al.224
	Pemetrexed	30%	Izzedine ²²⁵
	Tegafur = S-1	60%	Tomiyama et al.226
	Carboplatin	84%	Oguri et al.227
	Fludarabine	2F-Ara-A 25%	Kielstein et al.228
	Fluorouracil = 5-FU infusion	FBAL 60%	Rengelshausen et al.229
	Iodine 131	50%	Fofi et al.230
	Irinotecan	SN-38 50%	Koike et al.231
Aminoglycoside	Gentamicin	75%	Veinstein et al.203
	Tobramycin	80%	Kamel et al.177
Contrast agent	Gadolinium	65%–74%	Rodby ²²²

dFdU, 2',2'-Difluorodeoxyuridine; FBAL, α -fluoro- β -alanine; HD, hemodialysis; M, metabolite.

and account for changes among and within individuals. Therapeutic drug monitoring requires availability of rapid, specific, and reliable assays and known correlations of drug concentration to therapeutic and toxic outcomes. Indications for therapeutic drug monitoring include an experimentally determined relationship between plasma drug concentration and the pharmacologic effect, medications with a narrow therapeutic window, knowledge of the drug level influences management, potential patient compliance problems, and if the drug dose cannot be optimized by clinical observation alone. Examples of drugs analyzed by therapeutic drug monitoring include digoxin, lithium, phenytoin, theophylline, valproic acid, warfarin, tacrolimus, sirolimus, aminoglycosides, vancomycin, and clozapine. Unless therapeutic drug monitoring is being used to forecast a dose or there are concerns about toxicity, samples should be taken at steady state (four to five half-lives after starting therapy). The timing of the collection of the sample is important as the drug concentration changes during the dosing interval. The least variable point in the dosing interval to collect a blood sample is just before the next dose is due (trough level). The drug concentration is complementary to and not a substitute for clinical judgment, so it is important to treat the individual patient and not the laboratory value. Drug concentrations may be used as surrogates for drug effects, so therapeutic drug monitoring may assist with dose individualization. It can also be used to detect toxicity or nonadherence, so therapeutic drug monitoring can optimize patient management and improve clinical outcomes.¹⁷⁸

Hypoalbuminemia may influence interpretation of drug concentrations because the total drug concentration may be reduced, even when the active unbound drug concentration generally is not. Unbound drug concentrations are often not clinically available, so clinicians must empirically consider the influence of hypoalbuminemia in their interpretation of measured total drug concentrations, as in the case of phenytoin and several antibiotics (e.g., ceftriaxone and daptomycin).^{55,179,180}

DRUG DOSING IN PATIENTS WITH ACUTE KIDNEY INJURY

Critically ill patients frequently develop AKI ranging from 5% on admission to¹⁸¹ 50% in the ICU, as non-same-day hospitalization care is frequently complicated by AKI.⁹⁻¹³ In most cases, drug dosing is based on drug disposition information derived from studies in stable patients with CKD. There are large gaps in knowledge about drug metabolism and disposition in patients with AKI; thus patients may be at significant risk for underdosing and overdosing. Before the KDIGO consensus on AKI, more than 30 definitions of AKI were published in the literature.^{182,183} AKI may occur as part of multiorgan dysfunction in critically ill patients or as isolated AKI.¹⁸⁴ AKI-related, in-hospital mortality rates vary from up to 70% in ICU patients¹⁸⁵ to 35% in other hospitalized patients.¹⁸⁶

The potential effects of AKI on drug dosing are of major consequence because AKI patients are often critically ill and require multiple drug therapies, some of which may be nephrotoxic or require dose modification

in the setting of AKI. The pharmacokinetic changes in absorption, distribution, metabolism, and excretion are foundational to optimal patient care.^{7,51,135,187,188} The clinician must appreciate these factors and realize that they may worsen and improve over the period of evolution or recovery of the AKI episode. Critically ill patients with AKI typically have minimal oral intake of food and liquids and commonly require parenteral administration of drugs otherwise given orally (e.g., antihypertensives and immunosuppressives).

There is a paucity of dosing algorithms to guide pharmacotherapy, derived from investigations of the pharmacokinetics and pharmacodynamics of multiple-dose studies in patients with AKI, and much variability exists.¹⁶¹ Most of the critical care literature and almost all FDA or EMA product labeling contain drug dosage recommendations derived from observations of patients with CKD and ESKF. The limited data available in the setting of AKI have predominantly been developed by clinicians; rarely is this information incorporated into official product labeling. The principles of drug dosage regimen modification described earlier for use in CKD thus remain the foundation for therapy optimization in patients with AKI. For patients on CRRT, dosing recommendations can be found in the literature. Generally, the dose in CRRT corresponds to around an eGFR of 30 mL/min, which is higher than that in intermittent HD.

LOADING DOSE

Many patients with AKI exhibit an expanded extracellular fluid volume, and the distribution volume is much larger than under normal conditions. Thus the D_{start} may need to be higher than the normal starting dose for persons with normal kidney function. Because of the V_D of many drugs, especially hydrophilic antibiotics, including β -lactams, cephalosporins, and carbapenems, is significantly increased in the presence of AKI, the administration of proactive D_{start} (25% > normal) is highly recommended.

MAINTENANCE DOSE

Forecasting the degree and rate of change in kidney function and fluid volume status is extremely challenging. A higher-than-normal maintenance dose might be needed in not only overhydrated patients but also ICU patients with sepsis and temporary glomerular hyperfiltration. Hyperfiltration associates with augmented kidney clearance that may lead to drug under dosage.¹⁶⁰ Thus maintenance dosing regimens for many drugs, especially antimicrobial agents, should be initiated at supranormal or near-normal dosage regimens, and adjustments be made 3 days later based on the relationship between drug pharmacokinetic characteristics and kidney function. Prospective measurement of serum drug concentrations and analysis using state-of-the-art pharmacokinetic and pharmacodynamic approaches should be used whenever possible.

DRUG DOSING IN PATIENTS UNDERGOING HEMODIALYSIS

The optimization of pharmacotherapy for patients receiving maintenance HD and emergent HD are both critically dependent on the availability of reliable

information from well-designed pharmacokinetic studies.^{7,17,25,190-195} The impact of HD on drug dosing requirements is dependent on the drug characteristics and dialysis prescription. Drug-related factors include molecular weight (MW) or size, degree of protein binding, and distribution volume.^{135,192} The vast majority of HD filters used before the mid-1990s were generally impermeable to drugs with an MW > 1 kDa.¹⁹⁰⁻¹⁹² Dialysis membranes are now predominantly composed of semisynthetic or synthetic materials, which have larger pore sizes, and this allows the ready passage of drugs that have an MW up to 20 kDa (i.e., digoxin). For more on clearance of poisons with dialysis, see [Chapter 66](#).

Drug clearance during dialysis can occur by three different processes.^{7,192} Drug removal by conventional HD occurs primarily by diffusion down a concentration gradient from the plasma to the dialysate. Removal of low-MW drugs is enhanced by increasing blood and dialysate flow rates and by using large-surface-area dialyzers. Larger molecules require more porous membranes for increased removal. The clearance of a drug by conventional HD can be estimated from the unbound fraction (f_u) and the following relationship:

$$Cl_{HD} = f_u \cdot Cl_{urea} \cdot (60/MW_{drug}),$$

where Cl_{HD} is the drug's clearance by HD, Cl_{urea} is the dialyzer clearance of urea, and MW_{drug} is the MW of the drug. The urea clearance for most conventional dialyzers varies between 150 and 200 mL/min and is comparable with values reported with high-flux hemodialyzers.¹⁹² With high-flux HD, the V_D and degree of protein binding of the drug become more important determinants of dialyzer clearance.¹⁹⁶ The convective transport and removal of drugs during high-flux HD depends primarily on filtration pressure gradient, treatment time, blood, and dialysate flow rates. Despite the widespread adoption of high-flux HD in certain parts of the world, there are sparse quantitative data on drug clearance.

There has been little investigation about the effects of sustained low-efficiency hemodialysis (SLED) regimens on drug disposition or comparison among modalities.¹⁹⁷⁻²⁰⁰ Only a few agents have been evaluated during the delivery of one of these dialytic variants. Slow nocturnal dialysis required a significant increase in gentamicin dosage to achieve therapeutic levels compared with conventional thrice-weekly dialysis.^{197,198} The variability in drug clearance was high and did not correlate with small solute clearance. Similar findings were reported with cefazolin.¹⁹⁹ The cefazolin clearance during nocturnal HD was slightly lower ($Cl = 1.65$ L/hr) than during high-flux intermittent HD ($Cl = 1.85$ L/hr). However, a greater percentage of cefazolin was removed in 8 hours of nocturnal HD (80%) than conventional 4-hour high-flux HD (60%). The investigators concluded that a dosing regimen of a 2-g D_{start} followed by 1-g IV dose after each HD was sufficient to achieve concentrations $6 \times MIC$ for *Staphylococcus* species for at least 70% of the dosing interval. As a result, though drug dosing in patients receiving one of these dialytic variants may be empirically increased, drug dosing regimens should be guided by drug-level monitoring when feasible. Drugs with a molecular size of 500 to 5000 Da appear to be particularly likely to have an increased clearance with

nocturnal dialysis. Studies of modeled clearance have suggested that frequent HD regimens would be associated with enhanced clearance (and thereby the potential of under dosing) of daptomycin, for example.^{201,210} This enhanced clearance was confirmed in the setting of AKI when the pharmacokinetics associated with extended daily dialysis (EDD) was investigated.¹⁷⁹ These findings should be transferable to maintenance HD, with a degree of caution about the effects on distribution volumes that might arise in the setting of septic shock.²⁰³ One of the other effects of prolonged HD appears to be a reduction in rebound of drug concentrations after the termination of dialysis.^{204,205} The rebound is explained by the fact that the drug transfer from the peripheral to central compartment needs time rendering rapid intermittent HD is less efficient in clearing drugs than prolonged or continuous HD.

There are >100 different dialysis or hemofilters available, and at least four distinct variants of HD are currently being used.¹⁹² The effect of HD or hemofiltration on the disposition of a drug may vary markedly and, because dialyzer or hemofilter clearance is rarely evaluated more than once, clinicians have to extrapolate data from one procedure to another.²⁰⁶⁻²⁰⁸ The enhanced efficiency of 21st-century dialyzers mean that most of the literature for medications developed before 2000 probably reflects an underestimation of the impact of HD.¹⁹¹ Consequently, the dosages of some medications may need to be empirically increased by 25% to 50%. Therapeutic drug monitoring should be used for drugs with narrow therapeutic indices to optimize safety and efficacy.

ASSESSMENT OF THE IMPACT OF HEMODIALYSIS

The most common means for assessing the effect of HD is to calculate the dialyzer clearance of a drug ($Cl_{pharmacodynamic}$) from plasma, as follows:

$$Cl_D^p = Q_p \cdot ([A_p - V_p]/A_p),$$

where Q_p is plasma flow through the dialyzer, A_p is the concentration of drug in plasma going into the dialyzer, and V_p is the plasma concentration of drug leaving the dialyzer.^{190,205} This equation tends to underestimate HD clearance for drugs that readily partition into and out of erythrocytes. In addition, venous plasma concentrations may be artificially high if extensive ultrafiltration is performed, so $Cl_{pharmacodynamic}$ will be lower than it really is. Because of these limitations, the recovery clearance approach remains the benchmark for the determination of dialyzer clearance and can be calculated as follows^{163,190-192}:

$$Cl_D^r = R / AUC_{0-t},$$

where R is the total amount of drug recovered unchanged in the dialysate and AUC_{0-t} is the area under the predialyzer plasma concentration–time curve during the period that the dialysate was collected. The HD clearance values reported in the literature may vary significantly, depending on which of these methods were used.¹⁹¹ The least number of assumptions is needed when the fraction (FR) of the amount in the body is derived from half-life $T_{1/2}$ during time t on dialysis.

$$FR = 1 - \frac{0.693}{T_{1/2}} \cdot t$$

It is common practice in most HD units to administer drugs after dialysis to minimize the loss of drugs that would result from the additional clearance during HD. However, performing HD immediately after dosing has been proposed as an option for removal of toxic antibiotics^{175-177,203,208} and chemotherapeutic agents.²⁰⁹⁻²³¹ For anticancer drugs, the administration of a normal dose up to HD 2 to 12 hours before dialysis makes sense.²³² This strategy delivers the desired maximum plasma concentration effect while minimizing the toxic drug or metabolic²³² effects (see Table 56.7). Emerging pharmacokinetic and pharmacodynamic considerations suggest that administration after HD may not be the optimal approach for some antibacterial agents, such as aminoglycosides, colistin, and vancomycin.^{175-177,203,208} High-bolus dosing immediately before or during the last hour of dialysis has been proposed for some antibiotics, but there have been few clinical studies.

If the drug is given after dialysis, the postdialysis dose (D_{HD}) should first replace the amount eliminated during the interval between dialysis sessions (D_{fail}) that is the result of clearance by the patient's residual kidney function and nonkidney clearance, in addition, the fraction of drug removed by HD (FR) should be estimated to calculate the necessary supplementary dose (D_{suppl}). The dose the patient should receive after HD would thus be the sum of these two doses (see Fig. 56.5):

$$D_{HD} = D_{fail} + D_{suppl}$$

$$D_{HD} = D_{fail} + FR \cdot (D_{start} - D_{fail})$$

In practice, the dose after dialysis corresponds to the initial/starting dose D_{start} or slightly less ($D_{HD} \leq D_{start}$).

DRUG DOSING IN PATIENTS RECEIVING CONTINUOUS KIDNEY REPLACEMENT THERAPY

CKRT and hybrid KRTs are commonly used to manage patients with AKI and ESKF in ICUs.²³³ CKRT provides less of

a challenge for drug dosing than intermittent HD because its continuous nature is analogous with drug removal by native kidneys and potentially amenable to the use of standard, first-order drug clearance equations to calculate dosing. However, in practice, CKRT rarely proves as continuous as planned in unstable ICU patients. The CKRT modality and the delivered dialysis dose can also have significant effects on drug clearance. MW, membrane characteristics (highly variable between systems), blood flow rate, ultrafiltration rate, and dialysate flow rate determine the rate and extent of drug removal.²³⁴ Because most drugs have MWs < 1.5 kDa, drug removal by CRRT does not depend greatly on MW. The use of higher hemofiltration volumes, especially if infused prefilter, can also affect clearance. The removal of urea, creatinine, and vancomycin was increased by 15% to 25% by the predilution compared with postdilution modality.^{235,236}

CKRT clearances have been noted to decline over time because of accumulation of protein on the dialysis membrane over time. Clotting within the hemofilter's hollow fibers also reduces the overall surface area for clearance. Although these factors have received little direct investigation, it appears that they do affect drug clearance.^{237,238}

Drug protein binding also affects how much is removed during CKRT because only unbound drugs are available for elimination by CKRT. Protein binding of >80% provides a substantial barrier to drug removal by convection or diffusion. During continuous venovenous hemofiltration, drug clearance generally approximates the ultrafiltration rate. The addition of diffusion by continuous venovenous hemodiafiltration increases drug clearance and is dependent on the ultrafiltration and dialysate flow rates. As is the case during high-flux dialysis, drug removal often parallels the removal of creatinine. Thus the simplest method for estimating drug removal during CKRT is to estimate the total CrCl.^{7,237,238}

Hybrid KRTs, including sustained or slow low-efficiency dialysis (SLED), EDD, continuous SLED, slow low-efficiency daily dialysis, and slow low-efficiency daily hemodiafiltration, which use higher dialysate flow rates and shorter treatment periods (6–12 hours in duration), are frequently used.²³⁹⁻²⁴² As of 2020, hybrid KRT and slow low-efficiency dialysis pharmacokinetic data have been published for few drugs.^{6,162,243} The improvement of KRT machines and filters has rendered old dosing guidelines for drugs, especially antibiotics, obsolete and potentially hazardous because they are underdosed. Although there are only a few FDA or EMA official drug dosing recommendations for patients receiving CKRT, several published dosing guidelines are widely used.^{17,207,237,238,243} Unfortunately, these recommendations have generally not been prospectively evaluated and their influence on patient outcomes is largely unknown.

In the absence of FDA or EMA recommendations, tertiary reference sources, or any published studies relating to the handling of a drug by CKRT (common with agents that are new to the market), it may be necessary for the clinician to formulate a dosing regimen using the pharmacokinetics principles presented in this chapter. The lack of pharmacokinetics data for antibiotics during prolonged KRT and intermittent HD in critically ill patients has made the attainment and maintenance of effective concentrations

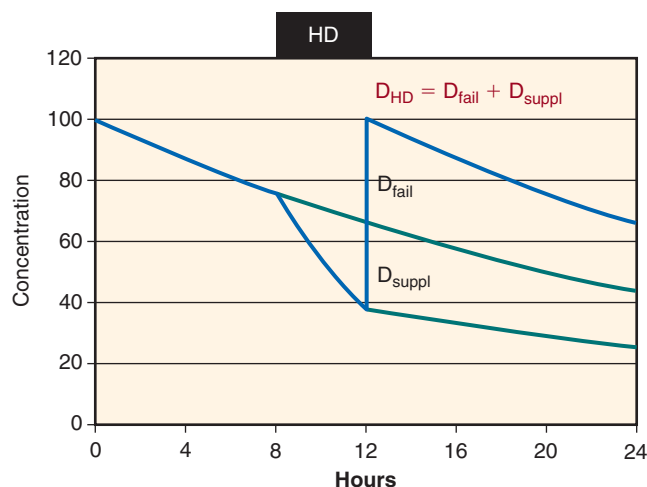


Fig. 56.5 Supplementary dose after hemodialysis (HD). To maintain therapeutic target concentrations, a supplementary dose must be given after hemodialysis to replace the removed fraction of the dose. The dose after dialysis (D_{HD}) combines both the adjusted maintenance dose (D_{fail}) and supplementary dose (D_{suppl}).

a major challenge.^{242,244} If the V_D is large (>1 L/kg), there is a low likelihood that CRRT will substantially remove the drug. The use of high-flux dialyzers or hemofilters allows for drugs with an MW < 20 kDa to be readily removed. If the clearance of the drug by CKRT or hybrid KRT is $<25\%$ of the patient's estimated total body clearance, a further dosing adjustment is probably unnecessary. By contrast, if CKRT or hybrid KRT results in an augmentation of drug clearance by 25% to 50%, an D_{start} based on the patient's estimated volume status should be given, and maintenance doses similar to those given to a patient with an eGFR of 30 to 50 mL/min can be used. Such estimates obviously must be supplemented by regular drug concentration measurements, where technically feasible.

DRUG DOSING IN PATIENTS UNDERGOING PERITONEAL DIALYSIS

Peritoneal dialysis, as practiced today, is unlikely to enhance total body clearance of any drug by >10 mL/min because most typical peritoneal dialysis prescriptions can achieve a urea clearance of about ≤ 10 mL/min. As most drugs are larger than urea, their clearance is even less; thus it is likely to be from 5 to 7.5 mL/min or less. Many studies performed in the 1970s and 1980s showed that drug clearances by peritoneal dialysis were in this low range, so one can conclude that peritoneal dialysis does not enhance drug removal to a degree that would require a special dosage regimen modification.²⁴⁵⁻²⁴⁷ Thus oral or IV drug therapy recommendations for patients with an eGFR <15 mL/min are likely clinically useful.

Intraperitoneal drug administration is well accepted for the treatment of peritoneal dialysis-associated peritonitis and other infections.^{247,248} Administration intervals depend on the half-life of the drug, which is mainly determined by residual kidney and nonrenal metabolic clearance. Long-standing experience with intermittent antibiotic administration in continuous ambulatory peritoneal dialysis (CAPD) exists for the glycopeptides vancomycin and teicoplanin, which can be administered at 5- to 7-day intervals, as well as for aminoglycosides and cephalosporins, which are suitable for once-daily dosing.^{249,250}

Patients treated by automated peritoneal dialysis (APD), with frequent short-dialysis cycles, may achieve higher plasma concentrations as compared with antibiotic loading in a single extended dwell period in patients on CAPD.²⁴⁹ Conversely, the higher dialysate flow and small-molecule clearance achieved with APD regimens may lead to a greater peritoneal clearance of antibiotics with the need for more frequent dosing.²⁴⁹ One common regimen involves adding antibiotics to each bag of dialysate solution used during APD exchanges. The choice of antibiotic, dosage, and frequency may vary depending on the patient's clinical condition, local guidelines, and presence of any allergies or drug interactions. The International Society of Peritoneal Dialysis (ISPD) guidelines emphasize the importance of individualizing antibiotic treatment based on factors such as the patient's medical history, previous infectious complications, and local microbiologic data.²⁴⁸

For intermittent maintenance dosing, a long nighttime dwell time should be used in patients on CAPD and a

long daytime dwell time in patients on APD. In clinical practice, intraperitoneal antibiotic dosing has not been unequivocally successful in eradicating bacterial growth, partially questioning the concept of antibiotic back diffusion into the peritoneal cavity. Thus antibiotics may need to be administered IV to enhance the antimicrobial effect.

Clinical Relevance

Hemodialysis removal of therapeutic drugs can dramatically impact the clinical outcomes of patients. High rates of removal can lead to therapeutic failures, morbidity, and even death. The advances in dialyzer membrane technology continue to increase the efficiency of the dialysis process, and thus therapeutic failures become more common, especially if clinicians base their prescribing practices on data derived from low-efficiency dialyzers that are no longer clinically used.

CLINICAL BOTTOM LINE

Recommendations for dosing of selected drugs in patients with CKD and AKI are given in Table 56.8. These are meant only as a guide and do not imply the safety or efficacy of a recommended dose in an individual patient. A D_{start} equivalent to the usual dose in patients with normal kidney function should be considered for drugs with half-lives >12 hours. No controlled clinical trials have established the efficacy of these dosage recommendations. The effect on drug removal of HD, CRRT, and ambulatory peritoneal dialysis is variable, and the values in the table are more qualitative than quantitative. Most of these recommendations were established before high-efficiency HD treatments were practical, continuous cycling nocturnal peritoneal dialysis was common, and diffusion was added to hemofiltration in CRRT. Consequently, therapeutic drug monitoring should be performed whenever possible, especially in patients where response to therapy may be suboptimal.

CONCLUSION

The adverse outcomes associated with inappropriate drug dosing in patients with kidney dysfunction have rarely been quantified but warrant future investigation. The utilization of FDA or EMA drug dosage recommendations in official prescribing information should be used for the initiation of therapy in most clinical situations. However, critically ill individuals likely have marked pharmacokinetic variability and may require the use of pharmacokinetic principles to determine the optimal drug dosage regimen and therapeutic drug monitoring to assess the impact.

Clinicians should therefore be aware of all the possible alterations in pharmacokinetics of drugs and what processes are likely to be altered in kidney failure. They should tailor drug therapy accordingly to ensure that patients with kidney dysfunction receive maximal benefits from their drug therapy while minimizing potential adverse outcomes.

Table 56.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury

Drug	Degree of Drug Dose Reduction or Interval Prolongation				Dosage Recommendations for Patients Receiving Kidney Replacement Therapy		
	GFR > 50 mL/min	GFR = 10–50 mL/min	GFR < 10 mL/min	HD	CAPD	CRRT	
Acetaminophen	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Acetazolamide	q4h	q6h	q8h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Acetohexamide	q6h	q12h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Acetohydroxamic acid	Avoid	Avoid	Avoid	Avoid	Avoid	Avoid	
Acetylsalicylic acid	100%	100%	Avoid	Unknown	Unknown	Unknown	
Acrivastine	q4h	q4–6h	Avoid	As normal GFR	As normal GFR	Dose as GFR 10–50	
Aceclovir	8 mg q6h	8 mg q8–12h	8 mg q12–24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Allopurinol	5 mg/kg q8h	5 mg/kg q12–24h	2.5–5 mg/kg q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Amantadine	100%	50%	33%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Amikacin ^a	q24h	q48–72h	q7days	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Amiloride	5–6 mg/kg q12h	3–4 mg/kg q24h	2 mg/kg q24–48h	5 mg/kg after HD	15–20 mg/L/day	7.5 mg/kg q24h	
Amiloride	100%	50%	Avoid	NA	NA	NA	
Amlodipine	100%	100%	100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR <10	
Amoxapine	100%	100%	100%	Unknown	Unknown	Dose as GFR 10–50	
Amphotericin B	q24h	q24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Amphotericin B lipid	q24h	q24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Ampicillin	250 mg–2 g q4–6h	250 mg–2 g q6h	250 mg–1 g q6h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Apixaban	100%	25–50 mL/min: 100% <25 mL/min: Not recommended	Not recommended	5 mg BID; 2.5 mg BID if age >79 years or body weight <61 kg	Unknown	Unknown	
Aripiprazole	100%	100%	100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR <10	
Atenolol	100% q24h	50% q24h	25% q24h	Dose as GFR 25–50	Dose as GFR <10	Dose as GFR 10–50	
Auranofin	6 mg q24h	3 mg q24h	Avoid	Avoid	Avoid	Dose as GFR 10–50	
Azathioprine	100%	75%–100%	50%–100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Aztreonam	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Benazepril	100%	50%–75%	25%–50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Bezafibrate	50%–100%	25%–50%	Avoid	200 mg q72h	200 mg q72h	200 mg q24–48h	
Bisoprolol	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Bleomycin	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Bretium	100%	25%–50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Bumetanide	100%	100%	100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR <10	
Bupropion	100% q24h	100% q24h	100% q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	
Butorphanol	100%	75%	50%	Unknown	Unknown	Dose as GFR 10–50	
Canagliflozin	100%	30–50 mL/min: Not Recommended <30 mL/min: Avoid	Avoid	Avoid	Unknown	Unknown	
Capreomycin	q24h	q24h	q48h	Dose as GFR <10	Dose as GFR <10	As normal GFR	
Captopril	100% q8–12h	75% q12–18h	50% q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50	

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Table 56.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury (Continued)

Drug	Degree of Drug Dose Reduction or Interval Prolongation			Dosage Recommendations for Patients Receiving Kidney Replacement Therapy		
	GFR > 50 mL/min	GFR = 10–50 mL/min	GFR < 10mL/min	HD	CAPD	CRRT
Carboplatin	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Carteolol	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Cefaclor	100%	100%	50%-100%	250-500mg q8h	250mg q8-12h	Dose as GFR 10-50
Cefadroxil	q12h	q12h	q24h	0.5-1.0g after HD	0.5g/day	Dose as GFR 10-50
Cefamandole	q6h	q6-8h	q8-12h	0.5-1.0g q12h	0.5-1.0g q12h	Dose as GFR 10-50
Cefazolin	q8h	q12h	50% q24-48h	15-20mg/kg after HD	Dose as GFR 10-50	Dose as GFR 10-50
Cefepime	q12h	50%-100% q24h	25%-50% q24h	Dose as GFR <10	Dose for GFR <10	1-2 g q12h
Cefixime	100%	75%-100%	50%	Dose as GFR <10	Dose for GFR <10	Dose as GFR 10-50
Cefotaxime	q6h	q6-12h	1 g q8-12h	Dose as GFR <10	Dose as GFR <10	1-2 g q12h
Cefotetan	q12h	q24h	q48h	1 g after HD	1 g q24h	Dose as GFR 10-50
Cefoxitin	q6-8h	q8-12h	q24-48h	1 g after HD	1 g q24h	Dose as GFR 10-50
Cefpodoxime	100%	100%	100-200mg q24-48h	Dose as GFR <10	Dose as GFR <10	As normal GFR
Cefprozil	100%	50% q12h	50% q12h	250mg after HD	Dose as GFR <10	Dose as GFR <10
Ceftazidime	100%	1-2g q24h	0.5-1 g q48h	1 g after HD	0.5-1 g q24h	1-2 g q12h
Ceftibuten	100%	50%	25%	400mg after HD	Dose as GFR <10	Dose as GFR 10-50
Ceftizoxime	q8h	q12h	q24h	1 g after HD	0.5-1.0g q24h	Dose as GFR 10-50
Cefuroxime (IV)	100% q8h	q8-12h	750mg q12h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Celiprolol	100%	100%	75%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Cephalexin	250-500mg q6h	250-500mg q8-12h	250-500mg q12-24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Cephadrine	100%	50%	25%	Dose as GFR <10	Dose for GFR <10	As normal GFR
Cetirizine	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Chloroquine	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Chlorpropamide	50%	Avoid	Avoid	Avoid	Avoid	Avoid
Chlorthalidone	q24h	Avoid	Avoid	Avoid	Avoid	Unknown
Cibenzoline	100% q12h	100% q12h	66% q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Cidofovir	50%-100%	Avoid	Avoid	No data	No data	Avoid
Cilazapril	75% q24h	50% q24-48h	10%-25% q72h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Cimetidine	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ciprofloxacin	100%	50%-100%	50%	250mg q12h	250mg q8h	200mg IV q12h
Cisplatin	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Clarithromycin	100%	75%	50%-75%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Clodronate	100%	50%	Avoid	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Clofazimine	100%	100%	100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Clofibrate	q6-12h	q12-18h	Avoid	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Clomipramine	100%	Start at lower dose, monitor effect	Start at lower dose, monitor effect	Dose as GFR 10-50	Dose as GFR 10-50	Dose as GFR 10-50
Clonidine	q12h	q12-24h	q24h	As normal GFR	As normal GFR	As normal GFR
Clopidogrel	100%	100%	100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Codeine	100%	75%	50%	As normal GFR	As normal GFR	As normal GFR
Colchicine	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Cyclophosphamide	100%	75%-100%	50%-75%	50%	75%	100%
Cycloserine	q12h	q12-24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50

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	100%	15-50 mL/min: 75 mg BID	Not Recommended	Not Recommended	Unknown	Unknown
Dabigatran	100%	15-50 mL/min: 75 mg BID	Not Recommended	Not Recommended	Unknown	Unknown
Dapone	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Daunorubicin	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Didanosine	50%-100%	33%-50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Diflunisal	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Digitoxin	100%	100%	50%-75%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Digoxina	100% q24h	25%-50% q24h	10%-25% q24-48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Disopyramide	q8h	q12h	q48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Dobutamine	100%	100%	100%	As normal GFR	As normal GFR	As normal GFR
Doxacurium	100%	50%	50%	Unknown	Unknown	Dose as GFR 10-50
Dyphylline	75%	50%	25%	Dose as GFR <10	Unknown	Dose as GFR 10-50
Emtricitabine	q24h	q48-72h	q96h	Dose as GFR <10	Unknown	Dose as GFR 10-50
Enalapril	100%	50%-100%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ertapenem	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Erythromycin	100%	100%	50%-75%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Ethambutol	q24h	q24-36h	q48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ethchlorvynol	100%	Avoid	Avoid	Dose as GFR <10	Dose as GFR <10	NA
Ethionamide	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ethosuximide	100%	100%	75%-100%	As normal GFR	As normal GFR	Dose as GFR 10-50
Etoposide	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Famciclovir	100%	q12-24h	50% q24-48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Famotidine	100%	50%	20 mg q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Fentanyl	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Fexofenadine	q12h	q12-24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Flecainide	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Fluconazole	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Flucytosine	50 mg/kg q12h	50 mg/kg q24h	50 mg/kg q24-48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Fludarabine	75%-100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Foscarnet	28 mg/kg/q8h	15 mg/kg/q8h	6 mg/kg/q8h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Fosinopril	100%	100%	75%-100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Gabapentin	300-600 mg q8h	200-700 mg q12h	100-300 mg q24h	LD: 300 mg MD: 100-300 mg q24h	As normal GFR	As normal GFR
Gallamine	75%	Avoid	Avoid	NA	NA	Avoid
Ganciclovir	2.5-5 mg/kg q12h	1.25-2.5 mg/kg q24h	1.25 mg/kg q24h	Dose as GFR <10	Dose as GFR <10	2.5 mg/kg q24h
Gemfibrozil	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Gentamicina	5-7 mg/kg/day	2-3 mg/kg/day by levels	2 mg/kg q48-72h by levels	3 mg/kg after HD by levels	3-4 mg/L/day by levels	Dose as GFR 10-50
Gliclazide	50%-100%	20-40 mg/day	20-40 mg/day	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Glipizide	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR <10
Guanadrel	q12h	q12-24h	q24-48h	Unknown	Unknown	Dose as GFR 10-50
Guanethidine	q24h	q24h	q24-36h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Hydralazine	q8h	q8h	q8-12h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Hydroxyurea	100%	50%	20%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Hydroxyzine	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Idarubicin	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ifosfamide	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Iloprost	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50

Table 56.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury (Continued)

Dosage Recommendations for Patients Receiving Kidney Replacement Therapy						
Drug	Degree of Drug Dose Reduction or Interval Prolongation			HD	CAPD	CRRT
	GFR > 50 mL/min	GFR = 10–50 mL/min	GFR < 10mL/min			
Imipenem	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Indapamide	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	NA
Indobufen	100%	50%	25%	Unknown	Unknown	Unknown
Isoniazid	100%	100%	75%-100%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Kanamycin	7.5mg/kg q12h	7.5mg/kg q24-72h	7.5mg/kg q48-72h	50% the normal dose	15-20mg/L/day	Dose as GFR 10-50
Ketorolac	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Lamivudine	100%	50-150mg q24h	25-50mg q24h	Dose as GFR <10	Dose as GFR <10	50mg q24h
Lenalidomide	100%	10mg q24h for 30-60 mL/min	Avoid	5 mg q24h after HD	Avoid	Avoid
		15 mg q48h for 10-29 mL/min				
Lepirudin	100%	25%-50%	Avoid	Avoid	Avoid	Avoid
Levofloxacin	100%	50%	25%-50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Lincomycin	q6h	q6-12h	q12-24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Lisinopril	100%	50%-75%	25%-50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Lithium carbonate ^a	100%	50%-75%	25%-50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Lomefloxacin	100%	50%-100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Lorazepam	q12h	q24h	q3-5days	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Melphalan	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Meperidine	100%	75%	50%	Avoid	Avoid	Avoid
Meprobamate	q6h	q9-12h	q12-18h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Meropenem	500mg-2 g q8h	500mg-1 g q12h	500 mg-1 g q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Metformin	100%	30-50mL/min: 25%-50% 10-29 mL/min: 25%	Avoid	Avoid	Avoid	Avoid
Methadone	100%	100%	50%-75%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Methotrexate	100%	50%	Avoid	Avoid	Avoid	Dose as GFR 10-50
Methyldopa	q8h	q8-12h	q12-24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Metoclopramide	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Metocurine	75%	50%	50%	Unknown	Unknown	Dose as GFR 10-50
Mexiletine	100%	100%	50%-75%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Midazolam	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Midodrine	5-10 mg q8h	5-10 mg q8h	2.5-10 mg q8h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Milrinone	100%	100%	50%-75%	No data	No data	Dose as GFR 10-50
Mitomycin C	100%	100%	75%	Dose as GFR <10	Dose as GFR <10	As normal GFR
Mivacurium	100%	50%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Morphine	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as <10
Mycophenolate mofetil	100%	50%-100%	50%-100%	Dose as GFR <10	Dose as GFR <10	As normal GFR
N-Acetylcysteine	100%	100%	75%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Nadolol	q24h	q24-48h	q48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Nalidixic acid	100%	Avoid	Avoid	Avoid	Avoid	Avoid
Neostigmine	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Netilmicin ^a	4-7.5mg/kg/day	3-7.5mg/kg/day	2 mg/kg Q24h	2 mg/kg after each HD	IV: 2mg/kg Q48h	Dose as GFR 10-50
Nicotinic acid	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50

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Nitroprusside	100%	100%	Avoid	Avoid	Avoid	Dose as GFR 10-50
Nitrofurantoin	100%	75%	25%-50%	Dose as GFR <10	Unknown	Unknown
Nizatidine	75%-100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Norfloxacin	q12h	q12-24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ofloxacin	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Oxcarbazepine	100%	75%-100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Pancuronium	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Paroxetine	100%	50%-75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
p-Aminosalicylic acid	100%	50%-75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Penicillamine	100%	Avoid	Avoid	Avoid	Avoid	Avoid
Penicillin G	100%	75%	20%-50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Pentamidine	q24h	q24h	q24-36h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Pentazocine	100%	75%	50%	Dose as GFR <10	Unknown	Dose as GFR 10-50
Pentopril	100%	50%-75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Pentoxifylline	q8-12h	q12-24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Perindopril	2 mg q24h	2 mg q24-48h	2 mg q48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Phenobarbital	q8-12h	q8-12h	q12-16h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Phenylbutazone	100%	50%	Avoid	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Pipecuronium	100%	50%	25%	Avoid	Avoid	Dose as GFR 10-50
Piperacillin	q6h	q6-12h	q12h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Plicamycin	100%	75%	50%	Unknown	Dose as GFR <10	Dose as GFR 10-50
Pomalidomide	100%	Avoid	Avoid	Unknown	Unknown	Unknown
Pravastatin	100%	30-50 mL/min: 100% 10-30 mL/min: 10 mg q24h	10 mg q24h	Dose as GFR <10	Unknown	Unknown
Pregabalin	100% q8-12h	q24h	25% q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Primidone	q12	q12-24h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Probenecid	100%	Avoid	Avoid	Avoid	Avoid	Avoid
Procainamide	q4h	q6-12h	q8-24h	Follow levels	Dose as GFR <10	Dose as GFR 10-50
Propoxyphene	100%	100%	Avoid	Avoid	Avoid	Avoid
Propylthiouracil	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Pyrazinamide	100%	100%	50%-100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Pyridostigmine	100%	35%	20%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Quinapril	100%	2.5-5 mg q24h	2.5 mg q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Quinine	q8h	q8-12h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ramipril	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ranitidine	100%	150 mg q24h	75 mg q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Ribavirin	100%	Avoid	Avoid	Avoid	Avoid	Avoid
Rifampin	100%	50%-100%	50%-100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Rivaroxaban	100%	15-50 mL/min 15 mg q24h	Avoid	Avoid	Avoid	Avoid
Simvastatin	100%	100%	5 mg q24h	Dose as GFR <10	As normal GFR	As normal GFR
Sitagliptin	100%	30-50 mL/min: 50% 10-30 mL/min: 25%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Sotalol	100%	25%-50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10-50
Spironolactone	100%	Usual dose q12-24h	Avoid	Avoid	Dose as GFR <10	Dose as GFR 10-50
Stavudine	100%	50% q12-24h	50% q24h	Avoid	Dose as GFR <10	Dose as GFR 10-50
Streptomycin ^a	q24h	q24-72h	q72-96h	Dose as GFR <10	20-40 mg/L/day	Dose as GFR 10-50
Streptozocin	100%	75%	50%	Unknown	Unknown	Unknown

Table 56.8 Recommendations for Dosing Selected Drugs in Patients With Chronic Kidney Disease or Acute Kidney Injury (Continued)

Drug	Degree of Drug Dose Reduction or Interval Prolongation				Dosage Recommendations for Patients Receiving Kidney Replacement Therapy	
	GFR > 50 mL/min	GFR = 10–50 mL/min	GFR < 10 mL/min	HD	CAPD	CRRT
Sulfamethoxazole	q12h	q18h	q24h	1 g after dialysis	1 g/day	Dose as GFR 10–50
Sulfipyrazole	100%	100%	Avoid	Avoid	Avoid	Dose as GFR 10–50
Sulfisoxazole	q6h	q8–12h	q12–24h	2 g after dialysis	3 g/day	NA
Sulindac	100%	50%–100%	50%–100%	Dose as GFR <10	Dose as GFR <10	Dose as GFR <10
Sulotroban	50%	30%	10%	Unknown	Unknown	Unknown
Tazobactam	100%	75%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Teicoplanin	q24h	q24–48h	q48–72h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Temocillin	q12–24h	q24h	q48h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Terbutaline	100%	50%	Avoid	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Tetracycline	100%	100%	50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Thiazides	100%	100%	Avoid	Dose as GFR <10	Dose as GFR <10	NA
Thiopental	100%	100%	75%	NA	NA	NA
Ticarcillin	50–75 mg/kg q6h	50–75 mg/kg q8h	50–75 mg/kg q12h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Tobramycina	5–7 mg/kg/day	2–3 mg/kg/day	2 mg/kg q48–72h	3 mg/kg after HD	3–4 mg/L/day	Dose as GFR 10–50
Tolvaptan	100%	100%	Avoid	Avoid	Avoid	Avoid
Topiramate	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Topotecan	75%	50%	25%	Dose as GFR <10	No data	No data
Tramadol	100%	50–100 mg q8h	50 mg q8h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Tranexamic acid	50%	25%	10%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Trazodone	100%	100%	Avoid/50%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Triamterene	100%	Avoid	Avoid	Avoid	Avoid	Avoid
Trimethoprim	q12h	q12h	q24h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Trimetrexate	100%	50%–100%	Avoid	No data	No data	Dose as GFR 10–50
Tubocurarine	75%	50%	Avoid	Unknown	Unknown	Dose as GFR 10–50
Valganciclovir	50%–100%	450 mg q24–48h	450 mg q72–96h	Avoid	Avoid	450 mg q48h
Vancomycin ^a	1 g q12–24h	1 g q24–96h	1 g q4–7days	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Venlafaxine	100%	25%–50%	25%–50%	Dose as GFR 10–50	Dose as GFR <10	Dose as GFR 10–50
Vigabatrin	100%	50%	25%	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Zalcitabine	100%	q12h	q24h	Dose as GFR <10	No data	Dose as GFR 10–50
Zidovudine (AZT)	100% q8h	100% q8h	50% q8h	Dose as GFR <10	Dose as GFR <10	Dose as GFR 10–50
Zileuton	100%	100%	100%	Dose as GFR <10	Unknown	Dose as GFR 10–50
Zopiclone	100%	3.75–5 mg daily	3.75–5 mg daily	Dose as GFR <10	Dose as GFR <10	Dose as GFR <10

^aAdjust dose to achieve desired serum concentrations using measured serum concentrations and pharmacokinetic modeling principles.

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QUESTIONS

- All of the following are potential mechanisms by which the volume of distribution of drugs is increased in patients with chronic kidney disease *except*:
 - Decreased plasma protein binding
 - Increased tissue binding
 - Increased fluid excretion
 - Increased fluid status

Answer: c

Rationale: The absorption, distribution, metabolism, and renal excretion of many drugs is altered by chronic kidney disease. The increase in volume of distribution (V_D) may be the result of pathophysiologic alterations in body composition, fluid overload secondary to excessive fluid administration or intake, decreased protein binding, or increased tissue binding.

- Renal clearance of a drug is a composite of all of the following *except*:
 - Absorption
 - Glomerular filtration rate
 - Reabsorption
 - Secretion

Answer: a

Rationale: Kidney function is the most quantifiable determinant of renal drug clearance. It is important to note that the term “kidney function” includes the combined processes of glomerular filtration, tubular secretion, and reabsorption, as well as endocrine and metabolic functions. Alterations in any or all of these functions secondary to chronic kidney disease may have a dramatic effect on drug clearance.

- Which of the following statements regarding the effects of dose and dosing interval adjustments in patients with reduced renal function is true?
 - If the dose is reduced while the dosing interval remains unchanged, the peak will be higher and the trough lower.
 - If the dose remains unchanged while the dosing interval is increased, the peak and trough concentrations in patients with reduced renal function will be similar to those in patients with normal renal function.
 - If the dose is reduced while the dosing interval remains unchanged, the peak and trough concentrations in patients with reduced renal function will be similar to those in patients with normal renal function.
 - If the dose remains unchanged while the dosing interval is increased, the peak will be higher and the trough lower.

Answer: b

Rationale: The initial dose for chronic kidney disease (CKD) patients should be the same as the dose recommended for those with normal renal function unless the drug's volume of distribution is known to be altered in the presence of CKD or a concomitant disease, in which case the dose should be increased proportionally. This is the case in order to achieve the same peak. For the similar trough concentrations to be achieved, the interval should

be extended as the concentration–time profile of drugs is altered in patients with CKD.

- A 75-year-old woman with known atrial fibrillation is admitted to the hospital because of epistaxis while on warfarin. She needs a nasal tamponade and, because of subsequent fever, administration of cefazolin intravenously. In addition, the warfarin will be replaced by rivaroxaban. Because the serum creatinine is 200 $\mu\text{mol/L}$ (2.26 mg/dL) and her body weight is 80 kg, her estimated creatinine clearance by Cockcroft–Gault equation is 30 mL/min . Which of the following decisions is wrong?
 - The cefazolin therapy was started with 2000 mg.
 - The cefazolin maintenance dose was reduced to 30% of the usual standard dose and the administration interval extended from 8 to 12 hours.
 - Rivaroxaban was started 3 days later with 20 mg/day .
 - The rivaroxaban anti-Xa activity was measured 5 days after anticoagulation.
 - Because the anti-Xa activity was $> 200 \text{ ng/mL}$, the rivaroxaban dose was reduced to 15 mg/day .

Answer: b

Rationale: Only the cefazolin maintenance dose should be reduced, but the dosing interval should not be extended. Normal cefazolin half-life is 2.2 hours, but with kidney failure it is 33 hours. Accordingly, the fraction eliminated by the kidneys, FE, is $1 - 2.2/33 = 0.93$. The dose adjustment factor, Q, can be calculated from the equation:

$$Q = 1 - (\text{FE} [1 - \text{KF}]),$$

where KF is the fractional kidney function for this patient, which is 0.25 (CrCl of 30 mL/min divided by normal CrCl of 120 mL/min). From this, the dose fraction, Q, is determined to be 0.3. Because cefazolin action is considered to be time dependent, the administration interval should not be extended in addition to the dose reduction. Otherwise, the trough levels could fall below the anti-infective threshold concentration.

- Which of the following drugs is most likely to be removed by high-flux hemodialysis?
 - Amlodipine [molecular weight (MW) = 567; volume of distribution (V_D) = 21.0 L/kg ; plasma protein binding $> 95\%$]
 - Atenolol (MW = 266; V_D = 1.1 L/kg ; plasma protein binding = 3%)
 - Clindamycin (MW = 476; V_D = 0.8 L/kg ; plasma protein binding = 94%)
 - Apixaban (MW = 460; V_D = 0.3 L/kg ; plasma protein binding = 87%)

Answer: b

Rationale: The impact of hemodialysis on a patient's drug therapy is dependent on several factors, including the physicochemical characteristics of the drug, the dialysis conditions, and the clinical situation for which dialysis is performed. Drug-related factors that affect dialyzability include the molecular weight or size, degree of protein binding, and V_D . Dialysis membranes in the 21st century are predominantly composed of semisynthetic or synthetic

materials (e.g., polysulfone, polymethylmethacrylate, and polyacrylonitrile). These high-flux dialysis membranes have larger pore sizes and more closely mimic the filtration characteristics of the human kidney. This allows the passage of most solutes, including drugs that have a molecular weight of $\leq 20,000$ Da. Drugs that are small but highly

protein bound (i.e., $>90\%$) are not well dialyzed because both the principal binding proteins, α_1 -acid glycoprotein and albumin, have a high MW. For example, the MW of albumin is 60,000 Da. Finally, those drugs that are widely distributed with $V_D > 2$ L/kg are poorly removed by hemodialysis.